

## Kinetic Analysis of Complex Chemical Activation and Unimolecular Dissociation Reactions using QRRK Theory and the Modified Strong Collision Approximation

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*Dedicated to Prof. Dr. Dr. h. c. mult. Jürgen Troe  
on the occasion of his 60th birthday*

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### ***Rate Coefficient Prediction / Chemical Activation / Kinetics***

A method to predict temperature and pressure-dependent rate coefficients for complex bimolecular chemical activation and unimolecular dissociation reactions is described. A three-frequency version of QRRK theory is developed and collisional stabilization is estimated using the modified strong-collision approximation. The methodology permits analysis of reaction systems with an arbitrary degree of complexity in terms of the number of isomer or product channels. Specification of both high and low pressure limits is also provided. The chemically activated reaction of vinyl radical with molecular oxygen is used to demonstrate the approach. Subsequent dissociation of the stabilized vinyl peroxy radical is used to illustrate prediction of dissociation rate coefficients. These calculations confirm earlier results that the vinoxy + O channel is dominant under combustion conditions. The results are also consistent with RRKM results using the same input conditions. This approach provides a means to provide reasonably accurate predictions of the rate coefficients that are required in many detailed mechanisms. The major advantage is the ability to provide reasonable estimates of rate coefficients for many complex systems where detailed information about the transition states is not available. It is also shown that a simpler 1-frequency model appears adequate for high temperature conditions.

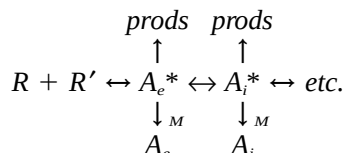
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## Introduction

Our interest is in chemically-activated, bimolecular reactions of the type,



where  $A_i^*$  is an activated complex and  $A_i$  is its stabilized adduct. The complex denoted by subscript  $e$  (entrance isomer) is that formed directly by the reactants. Subsequent isomerizations can lead to other complexes. These can either dissociate to form products or be collisionally deactivated to form stabilized adducts. The entrance isomer can also dissociate back to reactants, which can effectively be treated as another product channel. The overall bimolecular rates of reactants to either stabilized adducts or products can be written,

$$\frac{d[A_i]}{dt} = k^{stab}[R][R'] , \quad (1)$$

$$\frac{d[prod]}{dt} = k^{prod}[R][R'] . \quad (2)$$

The goal is to be able to calculate the rate constants  $k^{stab}$  and  $k^{prod}$  as functions of temperature, pressure and the collision parameters of the bath gas molecules. The limiting behavior at high and low pressures is also important. This information on the temperature and pressure dependence of rate coefficients can then be fit in a manner somewhat analogous to the implementation of fall-off reactions in CHEMKIN [1] to facilitate the incorporation of pressure-dependent chemical activation rate coefficients into detailed kinetic mechanisms. One such approach employing Chebyshev polynomials is described by Venkatesh *et al.* [2].

One method to describe such systems uses a QRRK calculation, based on a single geometric-mean frequency, to estimate the energy-dependent dissociation rate constants  $k(E)$  of the energized adducts. This is combined with a slightly simplified "modified strong collision approach" for collisional deactivation, and has been used to analyze a variety of gas phase, chemical activation radical reaction systems with good agreement to experimental data [3–7]. Other researchers [8, 9] have used the QRRK formalism with a more exact modified strong collision approach (as in this study) or used a QRRK formalism to compute  $k(E)$  [10, 11]. Craig *et al.* [10] note the suitability of this approach for explanation of product ratios in ion molecule reaction systems of reasonable size. Zhong and Bozzelli [12, 13] have recently utilized the method described in this work for analysis of cyclopentadiene and cyclopentadienyl radical reactions and compared results of pre-

dictions for benzene oxidation and pyrolysis, toluene oxidation, and phenol hydrogenation and oxidation to experiment. It was shown to yield reasonable results in these applications to complex chemical systems.

A potential shortcoming of the simple quantum Kassel approach is the use of a single geometric mean frequency to represent the  $3n-6$  vibration frequencies of the molecule. Weston [14] has made extensive comparisons of two quantum Kassel approaches (geometric and arithmetic mean), classical RRK theory, and a quantized treatment of classical RRK by Schranz *et al.* [15] with RRKM theory for a number of unimolecular dissociation reactions. He found that use of a geometric-mean Kassel approach tends to overestimate  $k(E)$ , the energy-dependent dissociation rate constant, for systems at energies within 10 kcal/mole of the critical energy for dissociation. This overestimate is worse when the arithmetic-mean frequency is used. Comparisons of  $k_0/k_\infty$  between RRKM and quantum Kassel theories in Weston's study showed that, while the results were reaction specific, overall the agreement for eight molecular systems analyzed was quite good. However, deviations of the quantum Kassel theory from RRKM occurred at the lower temperatures for several molecules. This is precisely the region where one would expect to find the largest differences, since the coarser energy graining with the single frequency will be most pronounced with the compressed low temperature thermal distribution function.

Use of QRRK theory is complementary to use of RRKM theory. With the RRKM approach, the A-factor for dissociation ( $k_\infty$ ) is computed from the properties of the transition state and the reactant, i.e., the vibrational and internal rotor frequencies, the state densities, and rotational parameters. The value of the resulting A-factor is often dominated by the low frequencies of the transition state and, for complex molecules, this structure is often sufficiently uncertain that adjustments are required to obtain reasonable values or agreement with experiment. Thus, although RRKM is more accurate in principle, the uncertainties in the input parameters may restrict its predictive value, unless significant effort is put into the assignments. While we encourage use of the more rigorous RRKM treatments where appropriate, this option is often not feasible for the modeler. For example, we find that more than 30 percent of the reactions in a typical combustion mechanism are either unimolecular dissociation or chemical-activated bimolecular reactions. Many of these are sufficiently complex that reliable assignments of frequencies and rotational parameters of the transition state at the required level of accuracy is not feasible. As the Arrhenius A-factors required for QRRK applications can be taken from the literature or approximated by analogy to similar reactions, there is significantly less opportunity (or temptation) to adjust the input parameters. The additional information required for the quantum Kassel reaction analysis is available from the literature, from thermodynamic parameter estimation codes [16], TST calculation codes [17], and transport property estimation methods [18].

Troe [19] points out a master equation approach is the best way to account for the competition between unimolecular reactions of the energized complex and its collisional deactivation. In practice, implementation requires detailed information about the magnitude and temperature dependence of the amount of energy transferred during collisions. Such information is not readily available for many systems of interest. Troe [22–24] has also developed the modified strong collision treatment to approximate collisional deactivation in thermally-activated systems. He has also developed improvements to the modified strong collision approximation to address chemical and photochemical activation [19]. We restrict ourselves to his modified strong collision approximation in this work, where the primary focus is calculation of the energy dependent rate coefficients of the energized adducts. We plan to address the master equation analysis in subsequent work.

This manuscript describes development of an improved three-frequency QRRK model [CHEMACT3F]. The three frequencies and the associated degeneracies are computed from the fits to heat capacity estimates [17]. These have been shown by Ritter [16] to accurately reproduce molecular heat capacities and vibrational state densities. This approach offers the advantage of avoiding the specification of the complete frequency distribution of the adduct; for many of the molecules of interest, such information is unavailable. Another improvement over the earlier CHEMACT code [5] is the ability to treat much more complex systems, e.g., systems with multiple stabilized adducts and multiple exit channels. Application of this improved approach will be illustrated by a detailed analysis of the vinyl + O<sub>2</sub> reaction, which has several stabilized adducts and multiple exit channels from the various adducts. In addition to its complexity, another reason for choosing this system is that detailed transition state structures are available and a multichannel RRKM analysis has been performed [20]. We were able to use the information to construct high-pressure rate constants (needed as input for the QRRK analysis) for each of the reactions in this system, thus allowing us to make an explicit comparison of the QRRK predictions to those obtained via the RRKM approach.

## Overview of the CHEMACT3F model

We consider the reactants to form the initial activated complex  $e$  with total forward rate constant  $k^{\text{imp}}$ . The activated molecules are differentiated according to their energy and the  $k_i(E)$  are calculated at each energy. The results are then summed over energy to obtain the total rate constants. We make the simplifying assumption that collisional stabilization can be approximated as a single step process using the modified strong collision approximation. Thus collisional energy redistribution need not be explicitly considered.

The differential concentration of activated complex  $i$  in the energy band  $E \pm \Delta E/2$  is denoted by  $AC_i(E)$ . Across the entire system,  $E$  is defined to a common zero, that corresponding to the entrance barrier height. At this point, no assumptions are made on the nature of  $\Delta E$ , except that the bands defined by  $\Delta E$  must be contiguous. The time rate of change of the activated population of  $i$  in each energy band can be written as the sum of incoming minus outgoing rates,

$$\frac{d[AC_i(E)]}{dt} = k^{inp}[R][R'] f(E, T) \delta_{i,e} + \sum_j (IS^T)_{i,j} AC_j(E) - AC_i(E) \left\{ \sum_j IS_{i,j} + \sum_m PD_{i,m} + k_i^s(T)[M] \right\}. \quad (3)$$

Here  $k_i^s(T)$  is the stabilization rate for the complex  $i$ ,  $[M]$  is the total concentration of bath species,  $IS_{i,j}$  is the isomerization rate matrix containing the differential rate constant from isomer  $i$  to  $j$  ( $IS^T$  denotes the transpose of this matrix), and  $PD_{i,m}$  is the product rate matrix containing the differential rate constant from isomer  $i$  to product channel  $m$ . The input channel is seen to enter the entrance well through the delta function  $\delta_{i,e}$  and  $f(E, T)$  is the product distribution resulting from this initial step. The sums over  $j$  are performed over all isomers while that over  $m$  is limited only to product channels connected to  $i$ .

Values for  $IS$  and  $PD$  are assigned through the differential rate functions,

$$IS_{i,j} = k_{ij}(E, \Delta E), \quad (4)$$

$$PD_{i,m} = k_{im}(E, \Delta E). \quad (5)$$

The input distribution function  $f(E)$  is derived through a detailed balance analysis of the reactant channel and entrance well [3]

$$f(E, T) = \frac{k_{er}(E, \Delta E) \varrho_e(E, \Delta E; T) \exp(-E/kT)}{\sum_E k_{er}(E, \Delta E) \varrho_e(E, \Delta E; T) \exp(-E/kT)}, \quad (6)$$

where  $k_{er}$  is the rate from the initial activated complex back to reactants and  $\varrho_e$  is the equilibrium distribution function describing the isomers over the entrance well.

By applying the steady state condition to Eq. (3), a matrix equation is formed for the normalized steady-state population vector  $AC_j^{ss}(E)/([R][R'])$ ,

$$M_{i,j} \left\{ \frac{AC_j^{ss}(E)}{[R][R']} \right\} = -\delta_{i,e} k^{inp} f(E, T), \quad (7)$$

where,

$$M_{i,j} = (IS^T)_{i,j} - \delta_{i,j} \left\{ \sum_k IS_{i,k} + \sum_m PD_{i,m} + k_i^s(T)[M] \right\}. \quad (8)$$

Once the steady state populations are obtained, the overall rate constants to products and stabilized adducts are found by summing the differential rate

constants over these distributions,

$$k_i^{stab} = \sum_E k_i^s(T)[M] \left\{ \frac{AC_i^{ss}(E)}{[R][R']} \right\}, \quad (9)$$

$$k_m^{prod} = \sum_E k_{im}(E, \Delta E) \left\{ \frac{AC_i^{ss}(E)}{[R][R']} \right\}. \quad (10)$$

### Expressions for the differential rate constant and distribution functions

In the formulation of the overall problem, an absolute energy scale is created onto which contiguous energy bands are defined as steps for the integration. This formalism allows an unambiguous specification of the interface between well systems. Calculations can be performed over each well in whatever local coordinates are convenient; all that is required is that the sum of all rates exiting a well in the specified energy band be passed to the analogous band in the next well. This flexibility allows for the mismatch of energy levels and spacings of the different isomers in the system.

In the simplest model, the energy levels of isomer *i* are described as a system of *s<sub>i</sub>* degenerate oscillators all at single frequency *ν<sub>i</sub>*. All energies are calculated with zero assigned to the bottom of the well. The index *n* enumerates the energy levels in increments of *hν<sub>i</sub>*, starting at *n* = 0. We ignore the zero-point energy since this is not available for distribution. The differential RRK reaction rate is derived using statistical arguments [21] and can be written in the form,

$$k_{im}(E, \Delta E) = A_{im} \frac{\sum_{n(\Delta E)} N(n - q_{im})}{\sum_{n(\Delta E)} N(n)}, \quad (11a)$$

$$N(k) = \frac{\Gamma(k + s_i)}{\Gamma(k + 1)\Gamma(s_i)}, \quad (11b)$$

where *A<sub>im</sub>* is the pre-exponential Arrhenius factor for channel *im*, and *q<sub>im</sub>* is the energy quanta corresponding to the barrier height *E<sub>im</sub>*. The ratio in Eq. (11a) corresponds to the number of states that can react divided by the total number of possible states (statistically, the number of reacting states are found by assuming *q<sub>im</sub>* of energy are already distributed in a single, specific fashion, and asking how many ways there are to distribute the remaining energy). The function *N(k)* gives the number of possible ways *k* quanta can be placed in *s* oscillators. Here, we have made the substitution [*Γ(k + 1) = k!*] to allow *q<sub>im</sub>* to assume non-integer values. To account for cases where more than one enumerated level can lie in the integration band, the total number of states contributing to both numerator and denominator

are counted by summing over all levels of  $n$  lying between  $E - dE/2$  and  $E + dE/2$ .

In the present formalism, potential complications would arise if an isomerization rate were to be calculated to a destination isomer that had no energy levels in the specified integration interval. To avoid these situations, the energy bands of the integration must be defined such as to include at least one level from each isomer in the system. This is easily done by letting  $\Delta E$  equal the largest energy level separation describing an isomer in the system.

The single-frequency RRK form of the equilibrium distribution function for isomer  $i$  can be expressed,

$$q_i(E, \Delta E; T) = [1 - \exp(-h\nu_i/kT)]^{s_i} \sum_{n(\Delta E)} N(n), \quad (12)$$

which is simply the number of states with energy  $E$  divided by the partition function. In the model used here, the  $(qk)$  product would be computed instead of  $q$  alone (taking the sum on  $n$ ), and the evaluation of a few common factors such as the partition function can be neglected since they are cancelled by the denominator of Eq. (6).

In the CHEMACT3F model, every isomer is modeled as having three distinct frequencies, each with a degeneracy factor. In this model, the computation of the differential rates is indexed to the smallest of the three frequencies. This links well to the present formalism since the sum of levels  $n$  can be taken over this single parameter and the additional degrees of freedom are handled implicitly in the three-frequency model itself. The three-frequency model can also be replaced with a generalized,  $n$ -frequency model (see Appendix B).

### Evaluation of the stabilization rate constant

In the modified strong collision approximation, collision stabilization is assumed to occur via a single step process in which a collision either stabilizes an activated complex completely or leaves it unchanged. The stabilization rate constant can then be defined in terms of a Lennard-Jones collision rate and a stabilization efficiency factor. Allowing the rate to be dependent on both the complex and the colliding species, this rate constant can be written

$$k_i^s(T) = \sum_c X_c \beta_c(T) Z_c(T), \quad (13)$$

where  $X_c$  is the mole fraction of collider  $c$ ,  $\beta_c(T)$  is the stabilization efficiency factor and  $Z_c(T)$  is the Lennard-Jones collision expression,

$$Z_c(T) = N_A \pi \sigma^2 \Omega^{[2,2]}(T) \sqrt{\frac{8kT}{\pi\mu}}. \quad (14)$$

Here  $N_A$  is Avogadro's number,  $\sigma$  is the collision diameter (average of that of collider and complex),  $\mu$  is the reduced mass of collider and complex

and  $\Omega^{(2,2)}(T)$  is given by the fit [18]

$$\Omega^{[2,2]}(T) = a_1(kT/\varepsilon)^{-c_1} + a_2 \exp(-c_2 kT/\varepsilon) + a_3 \exp(-c_3 kT/\varepsilon), \quad (15)$$

where  $a_1 = 1.16145$ ,  $a_2 = 0.52487$ ,  $a_3 = 2.16178$ ,  $c_1 = 0.14874$ ,  $c_2 = 0.77320$ ,  $c_3 = 2.43787$  and  $\varepsilon$  is the geometric mean  $[(\varepsilon_1 \varepsilon_2)^{1/2}]$  of the Lennard-Jones parameters of complex and collider.

### Calculation of the collision efficiency

For this analysis we choose to use the modified strong collision approach as a first approximation for estimation of fall-off in the unimolecular dissociation and chemical activation calculations. We note that Troe [19] has demonstrated that a master equation analysis is the preferred method for fall-off analysis, and he has also pointed out that the relation between average energy transfer and collision efficiency differs between thermally-activated and chemically-activated processes. The primary thrust of this research is, however, in methods to obtain estimates for  $k(E)$  in complex systems, and we therefore wish to defer treatment of the complexities of energy transfer. The collision efficiency  $\beta_c(T)$  is usually derived [22–24] in the unimolecular dissociation case. However, the same  $\beta_c$  must hold for both activation and deactivation through microscopic reversibility. The collision efficiency is found by comparing the results of the modified strong collision model with that obtained using a master equation analysis with an exponential form for the state-to-state collision probabilities. By equating the two solutions in the low pressure limit (and making a few additional assumptions), relatively simple expressions for  $\beta_c$  are found. Although these analyses were obtained in the case of a single exit channel, under the strong collision assumption the multiple exit case must yield identical result, with the substitution that the energy barrier in the solution is that which corresponds to the lowest exit channel for each well. (Only this exit channel is relevant because in the low pressure limit, all dissociating state are populated via a single-step collision from an energy level below the lowest energy barrier, since these are the only states with any significant population.)

The usual approximation [24] employed for  $\beta_c(T)$  is

$$\frac{\beta_c}{1 - \sqrt{\beta_c}} \cong \frac{-\langle \Delta E_c \rangle}{F_E kT}, \quad (16)$$

where  $\langle \Delta E_c \rangle$  is the average energy transferred in all collisions (and is a negative quantity) and  $F_E$  is a temperature-dependent integral involving the density of states  $\rho_{vib}(E)$  and is expressed,

$$F_E(T) = \frac{\int_{E_0}^{\infty} \rho_{vib}(E) \exp(-E/kT) dE}{kT \rho_{vib}(E_0) \exp(-E_0/kT)}. \quad (17)$$

Troe uses the Whitten-Rabinovitch approximation to calculate the density of states,

$$Q_{\text{vib}}(E) = \frac{[E_0 + a(E_0)E_z]^{s-1}}{(s-1)! \prod_{i=1}^s (h\nu_i)} \quad (18)$$

where  $s$  represents the number of vibrational modes,  $E_z$  is the zero point energy of the oscillator,  $E_0$  is the barrier height and  $a(E)$  is defined through the following:

$$\begin{aligned} a(E) &= 1 - \beta_v w & (19) \\ \log w &= -1.0506(E/E_z)^{0.25} \quad (E > E_z) \\ w^{-1} &= 5(E/E_z) + 2.73(E/E_z)^{0.5} + 3.51 \quad (E < E_z) \\ \beta_v &= (s-1) \frac{\sum_{i=1}^s \nu_i^2}{(\sum_{i=1}^s \nu_i)^2}. \end{aligned}$$

Substituting Eq. (18) into (17) and approximating  $a(E)$  by  $a(E_0)$ , Troe obtains an expression for  $F_E$ ,

$$F_E(T) = \sum_{i=0}^{s-1} \frac{(s-1)!}{(s-1-i)!} \left[ \frac{kT}{E_0 + a(E_0)E_z} \right]^i. \quad (20)$$

It should be noted that, although exact computational algorithms for  $Q_{\text{vib}}(E)$  are available [25], the present analysis is preferred because the analytic integral results in expressions that are easily evaluated.

$F_E$  is close to unity at low temperatures but exhibits an exponential-like dependence on temperature at high temperatures. For larger molecules the departure of  $F_E$  from unity starts at lower temperatures as more degrees of freedom are available. Eq. (16) suggests that  $\beta_c$  diminishes as  $F_E^{-1}$  in the limit that  $F_E$  becomes large. However Gilbert *et al.* [22] show that  $\beta_c$  levels off and even begins to turn around at high  $F_E$ . They recommend a higher-order approximation for  $\beta_c$  for use with large molecules.

Following Gilbert *et al.*, we replace Eq. (16) with

$$\beta_c = \left( \frac{\alpha_c}{\alpha_c + F_E kT} \right)^2 \frac{1}{\Delta}, \quad (21)$$

where  $\alpha_c$  represents the average energy of down-collisions and is related to  $\langle \Delta E_c \rangle$  though,

$$\langle \Delta E_c \rangle = \gamma_c - \alpha_c, \quad (22)$$

$$\gamma_c = \frac{\alpha_c F_E kT}{\alpha_c + F_E kT}, \quad (23)$$

where  $\gamma_c$  represents the average energy of up-collisions. Although not immediately obvious, Eqs. (21)–(23) are equivalent to Eq. (16) if  $\Delta$  is set to unity.

The factor  $\Delta$  in Eq. (21) can be decomposed into two terms,

$$\Delta = \Delta_1 - \left[ \frac{F_E kT}{\alpha_c + F_E kT} \right] \Delta_2 \quad (24)$$

where  $\Delta_1$  and  $\Delta_2$  are temperature-dependent integrals involving the density of states,

$$\Delta_1 = \frac{\int_0^{E_0} \rho_{vib}(E) \exp(-E/kT) dE}{\Delta_N} \quad (25)$$

$$\Delta_2 = \frac{\int_0^{E_0} \rho_{vib}(E) \exp(-E/kT) \exp[-(E_0 - E)/(F_E kT)] dE}{\Delta_N} \quad (26)$$

$$\Delta_N = \int_0^\infty \rho_{vib}(E) \exp(-E/kT) dE. \quad (27)$$

Again, using the Whitten-Rabinovitch approximation for  $\rho_{vib}$ , and approximating  $a(E)$  by  $a(E_0)$ , analytic expressions similar to Eq. (20) can be derived for  $\Delta_1$ ,  $\Delta_2$  and  $\Delta_N$ . This makes Eq. (24) ideally suitable for computation, especially since the factors  $\Delta_1$  and  $\Delta_2$  are independent of collider parameters.

One potential difficulty with using Eq. (21), is that the usual assumption of constant  $\langle \Delta E_c \rangle$  breaks down at high temperatures where  $F_E$  becomes large. At these conditions Eq. (23) forces  $\gamma_c$  and  $\alpha_c$  to approach one another in value. However, if the constant separation demanded by Eq. (22) is to be maintained,  $\gamma_c$  and  $\alpha_c$  must assume unreasonably large values. A more reasonable assumption is to fix  $\alpha_c$  as a temperature-independent constant, using Eqs. (22) and (23) to convert literature values of  $\langle \Delta E^c \rangle$  to  $\alpha_c$  at an appropriate temperature.

## Specification of the high and low pressure limits

In the low pressure limit, the population of activated complexes form a quasi-equilibrium distribution among themselves. Product rates are thus independent of pressure and stabilization rates are directly proportional to pressure. We can then define the low-pressure limiting rate constants by the equations,

$$\frac{d[A_i]}{dt} = k_{lp}^{stab} [R][R'][M] \quad (28)$$

$$\frac{d[prod]}{dt} = k_{lp}^{prod} [R][R'], \quad (29)$$

where the rate constants  $k_{lp}^{stab}$  and  $k_{lp}^{prod}$  are independent of pressure. The low

pressure case is computed by solving Eqs. (7) and (8) with the stabilization term  $\{k_i^s(T)[M]\}$  set to zero in Eq. (8).  $k_{ip}^{stab}$  and  $k_{ip}^{prod}$  are found using Eqs. (9) and (10), omitting the factor  $[M]$  in the expression of Eq. (9).

In the high pressure limit, the output channel for each isomer is dominated by the stabilization rate. In this case the following limiting forms are obtained:

$$\frac{d[A_i]}{dt} = \frac{k_{hp}^{stab}[R][R']}{[M]^{(d-1)}} \quad (30)$$

$$\frac{d[prod]}{dt} = \frac{k_{hp}^{prod}[R][R']}{[M]^d}, \quad (31)$$

where  $k_{hp}^{stab}$  and  $k_{hp}^{prod}$  are again pressure-independent. The form of these expressions depends on the position in the system occupied by the isomer from which the products originated. Here integer  $d$  represents the depth of the originating isomer relative to that of the entrance isomer, which is assigned a depth of 1 (i.e.,  $d$  equals 1 plus the minimum number of isomerizations required before the specified product channel can occur). Thus one consequence of a multiwell system is that the pressure dependence of the limiting high-pressure rate coefficients depend on the number of connected wells (isomers) and for complex systems the reaction order of pressure dependence can be much greater than unity.

To obtain solutions for the high pressure limit, Eq. (7) is solved with Eq. (8) replaced by,

$$(M_{hp})_{i,j} = ((IS_{hp}^T)_{i,j} - \delta_{i,j} k_i^s(T)). \quad (32)$$

Here the array elements of  $(IS_{hp}^T)_{i,j}$  are different than those used in Eq. (8). For each isomer  $i$ , only those input isomerization rates are counted that originate from an isomer  $j$  whose depth index  $d$  is one less than that of  $i$  (by definition, every isomer of depth  $d$  must be connected to at least one isomer of depth  $d-1$ , but not to any of lessor depth). All other rates in  $IS_{hp}^T$  are set to zero. Eqs. (9) and (10) are still used, except the factor  $[M]$  is again omitted in Eq. (9).

Having defined the high and low pressure limits, a generalization of the Lindemann form can be specified. This is done by postulating the simplest form which provides a smooth transition from one limit to another, with a shape factor introduced to account for any deviations;

$$Pr = \frac{k_{ip}[M]^d}{k_{hp}} \quad (33)$$

$$k^{stab} = \frac{k_{hp}^{stab}}{[M]^{(d-1)}} \left( \frac{Pr}{1 + Pr} \right) F \quad (34)$$

$$k^{prod} = \frac{k_{hp}^{prod}}{[M]^d} \left( \frac{Pr}{1 + Pr} \right) F. \quad (35)$$

Here  $F$  represents the equivalent Lindemann shape factor and  $Pr$  is the equivalent Lindemann reduced pressure. For further examination, the shape factors can be decomposed as [23]

$$-\ln F = L(T, Pr)[-\ln F_{\min}(T)], \quad (36)$$

where  $L(T, Pr)$  is a 'lineshape', normalized to unity at its maxima. As with the high-pressure limiting expressions, these rate coefficients can have a high reaction order with respect to pressure for complex multiwell systems, where  $d$  can be much larger than unity.

### Implementation of the model

The CHEMACT3F model is implemented as a Fortran 77 program which performs the calculation of all rates to products and stabilized adducts over a user specified temperature and pressure range. Eq. (7) is solved repeatedly, looping over energy, temperature and pressure. The structure of the code is arranged to optimize performance. The most computationally expensive part of the calculation is that of the rates, especially for the 3-frequency cases. Thus energy is taken as the outer loop and pressure, which involved the least computational overhead, as the inner loop. This arrangement has the added advantage that, although arrays are used to store intermediate variables vs. temperature, no storage is required based on energy. Thus there are no limitations imposed by memory on the integration step size. This ordering of computation does result in the artifact that all temperature and pressure cases are solved concurrently (with different convergence points on energy), with running sums over energy maintained for each temperature, pressure case throughout. Likewise, the denominator of  $f(E, T)$  is factored from the equations, also maintained as a running sum and incorporated as the final step. The high and low pressure limits are calculated as special cases within the pressure loop.

The sums over energy begin at zero (the input level) and proceed upwards to incorporate at least one band above the highest exit channel of the system. After this level is reached the sums are terminated (independently, for each temperature, pressure case) only when a convergence test on  $f(E, T)$  is satisfied. In the solution of the matrix equation of Eq. (7), provision has to be made for low energy cases where a high barrier to isomerization prevents any population from reaching a isomeric sub-system. Since these cases produce singular matrices, some logic had to be incorporated to identify inactive isomeric channels and remove the associated isomers from the computation prior to the matrix solver.

The treatment of  $\beta$  presented some difficulties. At large  $F_E$  the two terms in Eq. (24) become large and converge in value, resulting in a very small difference. Curiously over much of the high  $F_E$  range, the diminish-

ingly small  $\Delta$  in Eq. (21) is largely cancelled by the large  $F_E^2$  term, resulting in an almost constant  $\beta$  despite the severe variation with temperature of both these parameters. At large  $F_E$  though, effects of roundoff error become apparent. This problem was avoided by ‘freezing out’ the value of  $\beta$  at a temperature which corresponds to an  $F_E$  of  $10^6$ . The net effect on  $\beta$  is minimal as it is relatively invariant to temperature at these conditions. To prevent pathological cases,  $\beta$  is further constrained to be always less than unity.

Required program inputs include high-pressure rate constants (effective barrier heights and A-factors) for all product and isomerization channels, Arrhenius parameters for the high-pressure forward input rate (which is identical to the rate from reactants to the first stabilized adduct in the high pressure limit), frequency and collision parameters for all isomers, and collision parameters and mole fractions for all colliders. There are a number of straightforward techniques to obtain input parameters for CHEMACT3F without electronic structure calculations. Reactant and product energies and well depths can be taken directly from the thermochemical literature or estimated using group additivity methods. Reaction barriers can be estimated in the usual ways, e.g., by analogy to similar reactions. Rate constant values for the specific elementary reactions can be taken from databases on literature values or estimated using the principles of thermochemical kinetics. Additional details concerning the specification of input parameters can be found in Dean *et al.* [5]. For the single-frequency calculations, the geometric mean frequency of each isomer is used with a degeneracy equal to the total number of oscillators minus one half the number of hindered rotors. Frequencies and degeneracies for the three-frequency calculations are obtained using the procedures described by Bozzelli *et al.* [26].

### Example reaction system: $C_2H_3 + O_2$

To illustrate the capability of this code to handle complex systems, we used CHEMACT3F to predict temperature and pressure dependent rate coefficients for all the pathways of the chemically-activated reaction of vinyl radical with  $O_2$  as well as to predict dissociation rate coefficients of the stabilized adducts. A potential energy diagram for this system, based on Mebel *et al.* [20] is shown in Fig. 1. This system has 5 possible stabilized adducts and multiple exit channels. In addition, this system has been the subject of multiple experimental [27–29] and theoretical analyses [7, 20, 30–32]. A significant change since the earlier QRRK calculations [7, 30] was the recognition [31, 32] of the importance of formation of the 3-member ring dioxiranyl radical as opposed to the 4-member ring dioxetanyl radical used earlier. Mebel *et al.* [20] have analyzed this system using a multichannel RRKM approach, thus providing the means to directly compare the present results to theirs.

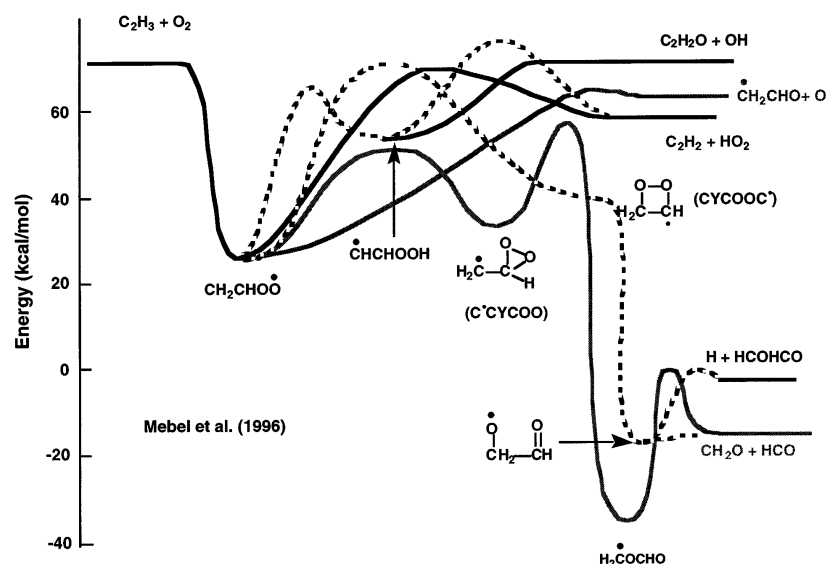


Fig. 1. Potential energy diagram, based upon Mebel *et al.* [20], for the addition of the vinyl radical to molecular oxygen.

As seen in Fig. 1, the initially formed vinyl peroxy adduct can react in many different ways. The lowest energy process is peroxy radical attack on the ipso carbon, forming a di-oxirane methylene radical, which, as reported by Carpenter [32], then isomerizes through an epoxide and cleaves the weak peroxy O–O bond. A further beta scission, preferring to maintain the ether linkage due to resonance or hyper-conjugation, opens this epoxide ring to form the  $\text{H}_2\text{C}^\bullet\text{--O--CHO}$  isomer. This isomer beta scissions to form  $\text{CH}_2\text{O} + \text{HCO}$  before stabilization can occur. This sequence, although the lowest in energy, starts with a relatively tight transition state to form the 3-member ring. A higher energy, but looser, transition state is associated with dissociation of the peroxy radical to form  $\text{C}^\bullet\text{H}_2\text{CHO}$  (vinoxy) + O. [Vinoxy is written in its more stable resonance form as a carbon-centered radical.] The energetics of this pathway are lower than usual for O–O bond breaking, since the vinoxy radical is resonantly stabilized. The peroxy radical can also undergo a concerted elimination to form acetylene +  $\text{HO}_2$ , isomerize via a hydrogen atom shift to form the vinyl hydroperoxide radical – which can then undergo beta-scission to form acetylene +  $\text{HO}_2$  (the same products formed via the concerted  $\text{HO}_2$  elimination from vinylperoxy), or isomerize to form a 4-member ring – which then can ring open to form product pairs such as  $\text{HCO} + \text{H}_2\text{CO}$  or  $\text{H} + \text{glyoxal}$ , or redissociate to reactants.

**Table 1.** Input parameters for the QRRK analysis of  $C_2H_3 + O_2 =$  products.

| High-pressure rate coefficients |                                                  |          |       |            |       |
|---------------------------------|--------------------------------------------------|----------|-------|------------|-------|
|                                 | Reaction*                                        | A        | n     | $\alpha$   | E     |
| $k_1$                           | $C_2H_3 + O_2 \Rightarrow CH_2CHOO^\bullet$      | 1.00E+13 | 0.0   | 0.0        | 0.0   |
| $k_{-1}$                        | $CH_2CHOO^\bullet \Rightarrow C_2H_3 + O_2$      | 1.61E+09 | 2.399 | 1.869E-03  | 44.2  |
| $k_2$                           | $CH_2CHOO^\bullet \Rightarrow C^\bullet CYCOO$   | 3.07E+07 | 1.811 | 6.108E-04  | 23.3  |
| $k_{-2}$                        | $C^\bullet CYCOO \Rightarrow CH_2CHOO^\bullet$   | 1.36E+06 | 2.206 | 8.807E-04  | 15.2  |
| $k_3$                           | $CH_2CHOO^\bullet \Rightarrow C^\bullet CHO + O$ | 1.16E+09 | 1.688 | 5.577E-04  | 37.7  |
| $k_4$                           | $CH_2CHOO^\bullet \Rightarrow C_2H_2 + HO_2$     | 6.19E+10 | 0.875 | 0.0        | 42.5  |
| $k_5$                           | $CH_2CHOO^\bullet \Rightarrow CH_2O + HCO$       | 3.34E+07 | 1.725 | 5.584E-04  | 43.6  |
| $k_6$                           | $CH_2CHOO^\bullet \Rightarrow C^\bullet HCHOOH$  | 1.37E+10 | 0.966 | 0.0        | 40.1  |
| $k_{-6}$                        | $C^\bullet HCHOOH \Rightarrow CH_2CHOO^\bullet$  | 7.06E+05 | 1.838 | 5.752E-04  | 13.6  |
| $k_7$                           | $C^\bullet HCHOOH \Rightarrow C_2H_2O + OH$      | 8.0E14   | 0.0   | 0.0        | 22.55 |
| $k_8$                           | $C^\bullet HCHOOH \Rightarrow C_2H_2 + HO_2$     | 3.32E+14 | 0.140 | -6.085E-04 | 26.8  |
| $k_9$                           | $C^\bullet CYCOO \Rightarrow H_2C^\bullet OCHO$  | 1.82E+06 | 2.199 | 8.576E-04  | 22.4  |
| $k_{-9}$                        | $H_2C^\bullet OCHO \Rightarrow C^\bullet CYCOO$  | 3.50E+05 | 1.879 | 6.135E-04  | 96.0  |
| $k_{10}$                        | $H_2C^\bullet OCHO \Rightarrow CH_2O + HCO$      | 9.66E+05 | 2.254 | 9.508E-04  | 30.1  |

\* Preexponential factor =  $A \cdot T^n \cdot \exp(-aT)$  ( $\text{sec}^{-1}$  or  $\text{cm}^3 \text{mole}^{-1} \text{sec}^{-1}$ )

Reaction barrier = E (in kcal/mole).

Notes:

Based on the results of Mebel *et al.* [20] (cf. text).

$k_1$  Adjusted to fit the low-pressure experimental data [28].

$k_5$  Assuming rate limiting step is formation of 4-member ring (CYCOOC $^\bullet$ ).

$k_7$  Est. A-factor for hydroperoxide [RO-OH] bond cleavage.

Adduct frequencies and degeneracies (freq in  $\text{cm}^{-1}$ , degen)

|                     |       |       |        |       |        |       |
|---------------------|-------|-------|--------|-------|--------|-------|
| $CH_2CHOO^\bullet$  | 452.6 | 5.05  | 1049.8 | 5.08  | 2915.9 | 4.31  |
| $C^\bullet HCHOOH$  | 481.5 | 7.31  | 1191.9 | 4.42  | 3113.3 | 2.27  |
| $C^\bullet CYCOO$   | 250.2 | 1.92  | 966.3  | 10.17 | 3208.0 | 2.27  |
| $H_2C^\bullet OCHO$ | 439.4 | 4.397 | 419.6  | 1.827 | 1808.6 | 7.202 |

Collision properties

$\sigma = 4.36 \text{ \AA}$   $\varepsilon/k = 451.0$   
Collision partner:  $N_2$  ( $\sigma = 3.62 \text{ \AA}$ ,  $\langle \Delta E_c \rangle = 830 \text{ cal/mole}$ )

## QRRK input data

The input parameters required for the QRRK analysis are listed in Table 1. (For simplicity, we did not include the reaction pathway to glyoxal + H, since inspection shows that this will be a minor contributor.) To facilitate comparison to the RRKM results of Mebel *et al.*, we have used their electronic structure calculation results for the energetics, geometries, and frequencies for intermediates, including both stabilized adducts and transition states, to compute the associated high-pressure thermal rate coefficients.

Table 2. Thermodynamic properties\*.

| Species                                        | $\Delta H^\circ(298)$<br>(kcal/mole) | $S(298)$<br>(cal/deg-mole) | $C_p$ (cal/deg-mole) |       |       |       |       |       |       |
|------------------------------------------------|--------------------------------------|----------------------------|----------------------|-------|-------|-------|-------|-------|-------|
|                                                |                                      |                            | 300                  | 400   | 500   | 600   | 800   | 1000  | 1500  |
| O                                              | 59.55                                | 38.47                      | 5.23                 | 5.14  | 5.08  | 5.04  | 5.01  | 5.01  | 4.98  |
| O <sub>2</sub>                                 | .00                                  | 49.01                      | 7.02                 | 7.02  | 7.23  | 7.44  | 7.65  | 8.04  | 8.35  |
| H                                              | 52.10                                | 27.39                      | 4.97                 | 4.97  | 4.97  | 4.97  | 4.97  | 4.97  | 4.97  |
| OH                                             | 9.45                                 | 43.88                      | 7.16                 | 7.08  | 7.05  | 7.05  | 7.15  | 7.33  | 7.87  |
| HO <sub>2</sub>                                | 3.80                                 | 54.73                      | 8.37                 | 8.95  | 9.48  | 9.96  | 10.78 | 11.43 | 12.47 |
| HCO                                            | 10.90                                | 53.42                      | 7.64                 | 7.95  | 8.50  | 9.10  | 10.21 | 11.08 |       |
| CH <sub>2</sub> O                              | -26.00                               | 50.92                      | 8.47                 | 9.38  | 10.46 | 11.52 | 13.37 | 14.81 |       |
| C <sub>2</sub> H <sub>2</sub>                  | 54.19                                | 48.07                      | 10.57                | 12.06 | 13.11 | 13.93 | 15.24 | 16.32 | 18.29 |
| C <sub>2</sub> H <sub>3</sub>                  | 71.62                                | 56.61                      | 10.01                | 11.97 | 13.66 | 15.08 | 17.32 | 19.05 | 21.85 |
| C*CHO                                          | 3.12                                 | 60.40                      | 12.92                | 15.31 | 17.44 | 19.24 | 22.10 | 24.12 |       |
| CH <sub>3</sub> CHOO*                          | 24.52                                | 68.03                      | 15.64                | 18.89 | 21.62 | 23.83 | 27.08 | 29.38 | 32.86 |
| C*HCHOH                                        | 49.85                                | 74.52                      | 18.37                | 21.66 | 23.91 | 25.74 | 28.23 | 29.84 | 32.97 |
| H <sub>2</sub> C*OCHO                          | -28.10                               | 69.36                      | 17.11                | 20.04 | 22.48 | 24.46 | 27.50 | 29.70 | 33.09 |
| C*CYCOO                                        | 32.09                                | 68.37                      | 16.38                | 19.73 | 22.45 | 24.61 | 27.76 | 29.92 | 33.26 |
| CYCOOC*                                        | 42.49                                | 71.32                      | 13.96                | 17.62 | 20.75 | 23.25 | 26.87 | 29.34 | 32.96 |
| Transition states:                             |                                      |                            |                      |       |       |       |       |       |       |
| From CH <sub>3</sub> CHOO* to:                 |                                      |                            |                      |       |       |       |       |       |       |
| C*CYCOO                                        | 48.67                                | 66.38                      | 14.68                | 17.85 | 20.47 | 22.56 | 25.62 | 27.78 | 31.08 |
| C <sub>2</sub> H <sub>2</sub> +HO <sub>2</sub> | 66.79                                | 67.96                      | 16.56                | 19.83 | 22.33 | 24.27 | 27.05 | 28.96 | 31.81 |
| C*HCHOH                                        | 64.77                                | 67.05                      | 14.86                | 18.47 | 21.35 | 23.57 | 26.72 | 28.82 | 31.83 |
| C*CHO+O                                        | 62.77                                | 71.37                      | 16.07                | 18.84 | 21.19 | 23.11 | 25.98 | 28.05 | 31.25 |
| CYCOOC*                                        | 68.95                                | 65.52                      | 14.11                | 17.42 | 21.59 | 22.34 | 25.52 | 27.74 | 31.81 |
| From C*CYCOO to:                               |                                      |                            |                      |       |       |       |       |       |       |
| H <sub>2</sub> C*OCHO                          | 55.84                                | 66.79                      | 15.01                | 18.05 | 20.61 | 22.67 | 25.72 | 27.88 | 31.17 |
| From C*HCHOH to:                               |                                      |                            |                      |       |       |       |       |       |       |
| C <sub>2</sub> H <sub>2</sub> +HO <sub>2</sub> | 75.52                                | 81.04                      | 20.71                | 23.47 | 25.58 | 27.23 | 29.64 | 31.35 | 33.94 |
| From H <sub>2</sub> C*OCHO to:                 |                                      |                            |                      |       |       |       |       |       |       |
| CH <sub>2</sub> O+HCO                          | -8.50                                | 72.93                      | 15.86                | 18.31 | 20.39 | 22.14 | 24.94 | 27.00 | 30.27 |

\* Standard state is 1 atm, 298 K.

First, these properties were used to compute the associated thermodynamic enthalpies and entropies using the usual statistical thermodynamic relations. These values are listed in Table 2. These in turn were used to compute the Transition State Theory rate coefficients. These rate coefficients were then fit to a rate expression of the following form:

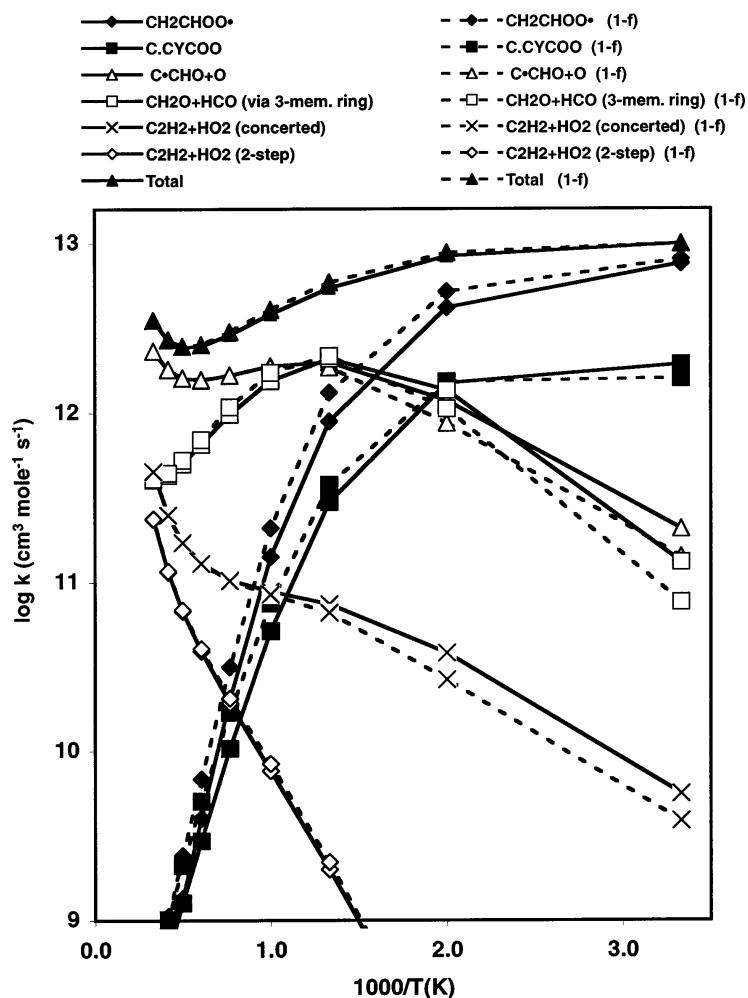
$$k_{\text{TST}} = A T^n \exp(-\alpha T) \exp(-E/RT) . \quad (37)$$

The reaction barrier  $E$  was fixed at the value obtained by Mebel *et al.*, for the energetics of the reaction being considered, specifically  $\Delta H(1000\text{K})$ . Although one could obtain a better fit by varying  $E$ , such a procedure would alter the energetics used for the QRRK calculations, and thus distort the results. The complex form for the pre-exponential was used to improve the accuracy of the fit.

The 3 frequencies and the associated degeneracies were obtained from fits to the heat-capacity as a function of temperature [26]. The fitting procedure allows non-integer degeneracies to better capture the temperature dependence of the heat capacity. An advantage of this approach is that heat capacities, even for reactive intermediates, can be accurately estimated using group additivity – thus allowing for the straightforward calculation of the necessary frequencies and degeneracies.

### CHEMACT3F results

The predictions for the rate coefficients from 300–3000 K at a pressure of 1 atm  $\text{N}_2$  for the major channels resulting from vinyl addition to oxygen are plotted in Fig. 2. The dominant product at low  $T$  is the stabilized peroxy adduct, with the stabilized 3-membered cyclic isomer the next most important product. Even though the barrier to dissociation of the peroxy radical to vinoxy ( $\text{C}^*\text{CHO}$ ) + O is also below the entrance channel, the total energy in the energized adduct is sufficiently low that the time required for dissociation is longer than that required for stabilizing collisions. Formation of the stabilized cyclic adduct occurs since the isomerization barrier here is substantially lower than that for dissociation to vinoxy. The lower barrier more than offsets the lower A-factor associated with the tighter isomerization transition state. However, the subsequent barrier for formation of  $\text{CH}_2\text{O}$  + HCO is sufficiently high to inhibit formation of these products. Of course, stabilization will scale with pressure, so one expects the exit channels to be much more important at lower pressures, as will be discussed in more detail later. As the temperature (and thus the internal energy of the energized adduct) increases, the exit channels (both the vinoxy + O as well as  $\text{CH}_2\text{O}$  + HCO) become more accessible. Above  $\sim 600$  K, these pathways dominate, with the vinoxy becoming the most important above  $\sim 1000$  K. Formation of acetylene via the concerted elimination of  $\text{HO}_2$  from the peroxy adduct has a lower barrier than the sequential reactions to these same prod-



**Fig. 2.** Predicted CHEMACT3F results, at 1 atm  $\text{N}_2$ , for the effect of temperature on the rate coefficients for formation of the major products resulting from the addition of vinyl radical to molecular oxygen. Also included are the results obtained using only a single geometric mean frequency.

ucts. Thus the concerted pathway is more important at low temperatures. However, the sequential pathways become competitive at the highest temperatures. These calculations reinforce the conclusions of our earlier work [7] that the vinoxy + O channel dominates at higher temperatures and should be included in combustion mechanisms.

As discussed earlier, one of the improvements incorporated into CHEM-ACT3F was the use of a three-frequency model to characterize the vibrational energy distribution. Fig. 2 also shows a comparison of the results obtained with the three-frequency model with those obtained with the simpler one-frequency (geometric mean) model. The results are surprisingly consistent, with the biggest deviations observed at the lower temperatures, as expected. We have observed similar behavior in comparison of other systems.

It is important to recognize that the rate coefficients presented in Fig. 2 represent the apparent rate of production of both stabilized adducts and final products. As will be discussed in more detail later, it is essential to also include the subsequent thermal activation of the stabilized adducts in any kinetic scheme. Without such reactions, the formation rate of stabilized adducts could be greatly overestimated, especially at higher temperatures and for adducts with relatively shallow wells.

Fig. 3 shows the calculated pressure dependence of the rate coefficients at a fixed temperature of 2000 K. For completeness, all of the possible products are included in this plot. Note that all of the exit channels have virtually no pressure dependence, even at pressure as high as 100 atm. At this high temperature, the internal energy in the adducts is sufficient to drive the dissociation pathways prior to collisional stabilization. At this temperature representative of that occurring in flames, the dominant pathway is formation of vinoxy + oxygen atoms. Since vinoxy is expected to rapidly beta-scission to form ketene + H, this pathway is effectively chain-branching, making the vinyl reaction an especially important one in combustion kinetics. Of the two pathways to formaldehyde + formyl, the route via the 3-membered ring adduct is dominant, although there is a small contribution from the path via the higher-energy 4-membered ring isomer. At this temperature, acetylene production via the concerted elimination is slightly favored over the sequential pathway. It is interesting to note that the formation of the diradical  $C_2H_2O + OH$  begins to become important at these higher temperatures. For completeness, this pathway should be included into reaction mechanisms. ( $C_2H_2O$  can beta-scission to form the ketyl radical + H.) Even at the highest pressures, the stabilization channels are only beginning to become significant. The greater rate coefficients for the vinyl peroxy adduct and the 3-membered ring isomer naturally follow from the relatively deeper wells for these two species. (Although the  $H_2C^{\bullet}OCHO$  well is  $\sim 30$  kcal/mole, the adducts above this well must enter with over 90 kcal/mole of internal energy, leading to little probability of stabilization prior to dissociation.)

Fig. 4 compares the predictions for the overall rate coefficient at low pressures to those measured by Knyazev and Slagle [27]. Predictions are shown for pressures of both 0.001 and 0.01 atm, spanning the experimental range of pressures. There is virtually no pressure dependence predicted,

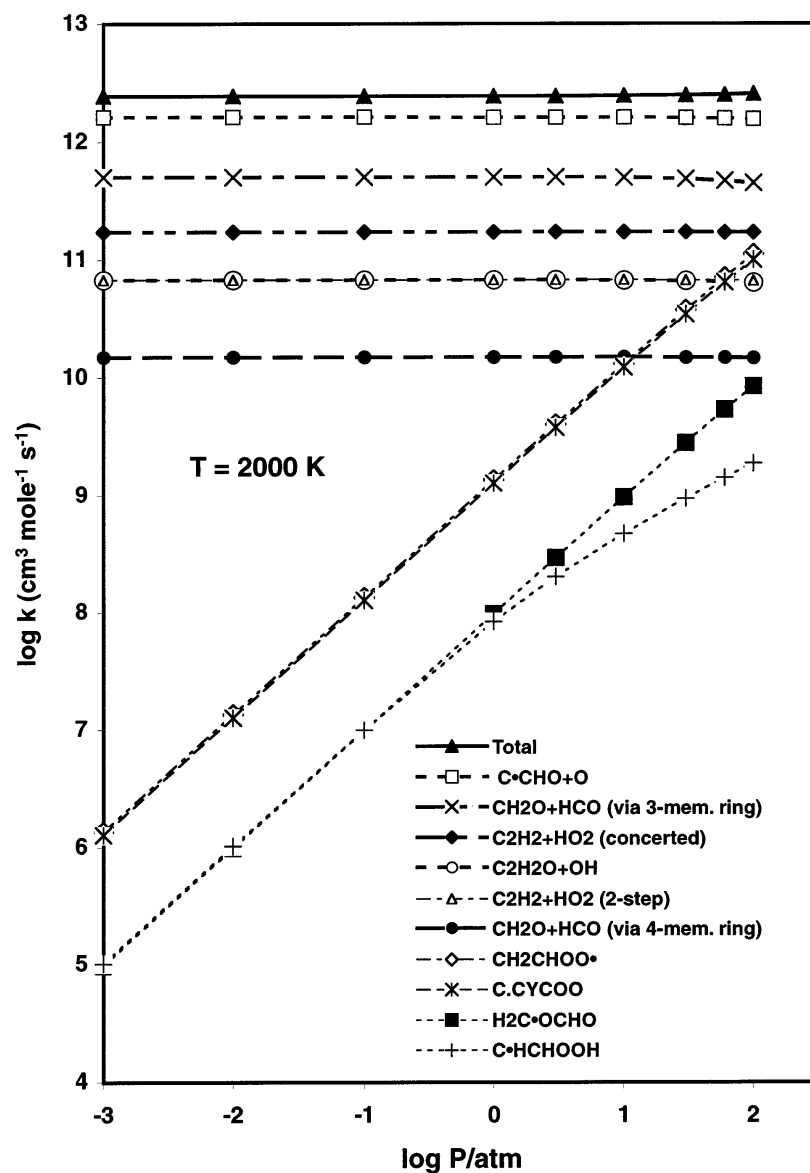
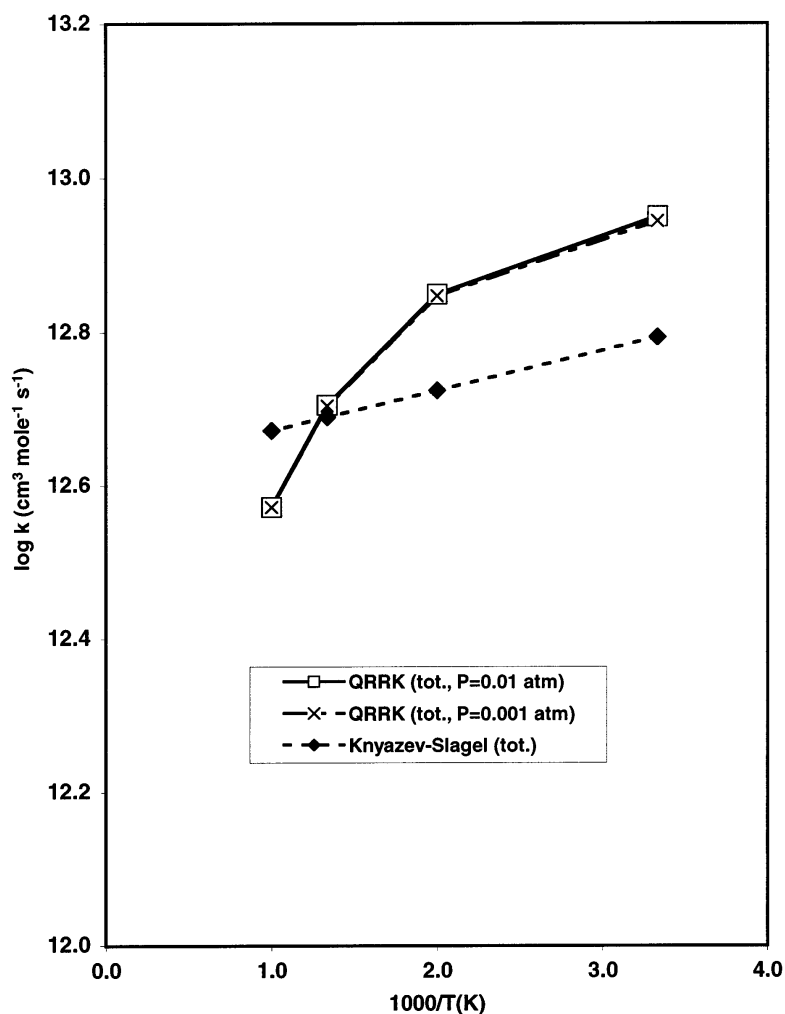
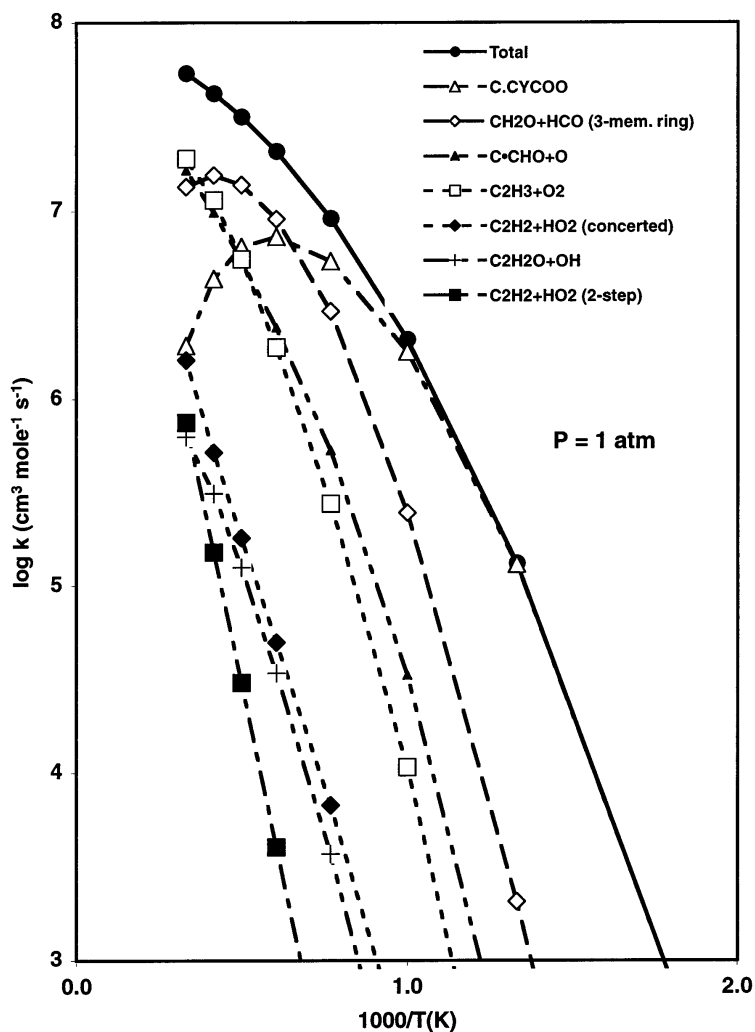


Fig. 3. Predicted CHEMACT3F results, at T = 2000 K, for the effect of pressure (with N<sub>2</sub> as the collision partner) on the rate coefficients for formation of all the products resulting from the addition of vinyl radical to molecular oxygen.



**Fig. 4.** Comparison of the predicted CHEMACT3F results to that measured by Knyazev and Slagle [27] for the temperature dependence of the total rate coefficient of the reaction between vinyl radical and molecular oxygen at 0.01 and 0.001 atm with He as the collision partner.

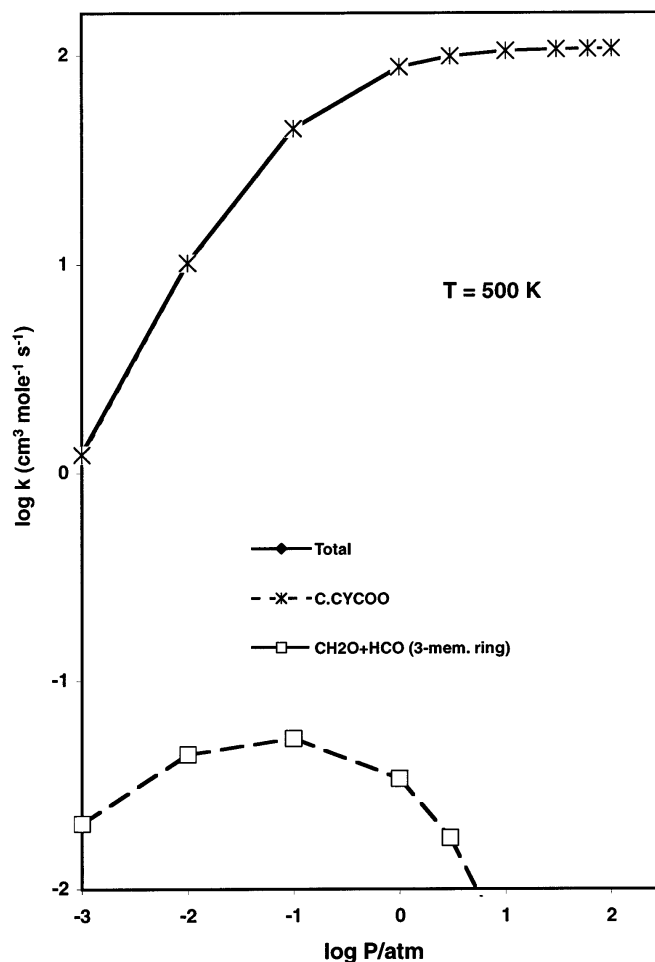
consistent with the experimental observations. Although the predictions show a slightly steeper decline in rate coefficient with temperature than observed, the overall agreement is encouraging. (This falloff in rate coefficient is due to increased dissociation of the adduct back out the exit channel as the temperature is increased.) The calculations show  $\text{CH}_2\text{O} + \text{HCO}$



**Fig. 5.** Predicted CHEMACT3F results, at 1 atm  $N_2$ , for the effect of temperature on the rate coefficients for the major product channels resulting from dissociation of the stabilized vinyl peroxy radical.

as the major product ( $\sim 60\%$ ) at 300 K, again consistent with experiment. With the lack of stabilization at the low pressures, the rate coefficients to both major exit channels are substantially higher than at 1 atm.

Given that stabilization is a major pathway under low temperature, higher pressure conditions, it is important to account for the subsequent reac-



**Fig. 6.** Predicted CHEMACT3F results, at  $T = 500$  K, for the effect of pressure (with  $N_2$  as the collision partner) on the rate coefficients for formation of the major product channels resulting from dissociation of the stabilized vinyl peroxy adduct.

tions of the stabilized adducts. As described in Appendix A, CHEMACT3F can also be used to calculate the temperature and pressure dependent dissociation rate coefficients of these adducts. Such predictions, for the most important adduct, vinyl peroxy, are shown in Fig. 5 for  $P = 1$  atm. At lower temperatures, the only product formed is the 3-membered ring isomer, since this is the lowest energy exit channel. The next channel to open up as the temperature increases is further reaction of this adduct to form  $CH_2O +$

HCO. This is followed by the vinoxy + O channel, and then the entrance channel  $\text{C}_2\text{H}_3 + \text{O}_2$ , again in order of increasing energy. Fig. 6 illustrates the predicted pressure dependence for vinyl peroxy dissociation at 500 K, a temperature where formation of this adduct is still quite important. The only pathway accessible is isomerization to the 3-membered ring adduct. For typical conditions of interest in modeling, this reaction is in the falloff regime.

Table 3 lists the parameters for non-Arrhenius fits to the rate coefficients for each channels at a range of pressures and temperatures.

### Comparison to RRKM results

Fig. 7 compares the QRRK results to the RRKM results of Mebel *et al.* [20] for  $P = 1 \text{ atm N}_2$ . Both calculations use the modified strong collision assumption, but the details of the collisional stabilization model used by Mebel *et al.* were not provided. The QRRK results yield approximately twice the stabilization rate as obtained by Mebel *et al.* at 300 K, with better agreement at higher temperature. The difference in stabilization is not unexpected, given the presumed differences in the calculation of the collision efficiency  $\beta$ . The increased QRRK stabilization at low temperature naturally lowers the rate coefficient for the non-stabilized channels. At higher temperatures, where the importance of stabilization diminishes, the QRRK and RRKM results exhibit reasonable agreement for all the channels.

Given the differences in predicted stabilization rates, a more direct comparison of the QRRK vs. RRKM approaches can be obtained by comparing the predicted branching ratios for the two methods. Fig. 8 compares the QRRK results, using both the three frequency and geometric mean models, to the RRKM results of Mebel *et al.* [20] for the branching ratios to the major products at high temperatures. We restrict the comparison to higher temperatures, since collisional stabilization is not important there. Over an extended temperature range, the various approaches give very similar results. As expected, the single frequency results merge into the 3-frequency results at the higher temperatures. The agreement between the QRRK and RRKM results are gratifying, suggesting that use of the three-frequency distribution (or even a 1-frequency at high temperatures) is adequate to characterize the overall vibrational frequency distribution. As discussed earlier, this approximation is especially useful for those complex systems where the detailed vibrational distribution — especially the hard-to-characterize low frequency torsions and internal rotations — are not known. It appears that, for such cases, use of a frequency distribution extracted from readily estimated heat capacities is adequate to characterize chemically-activated reactions.

**Table 3.** Calculated rate coefficients for the  $C_2H_3 + O_2$  system.  $k = AT^n \exp(-E/RT)$ , ( $\text{sec}^{-1}$  or  $\text{cm}^3 \text{mole}^{-1} \text{sec}^{-1}$ ,  $E$  in  $\text{cal/mole}$ ).

| Reaction                                   | A          | n     | E       | P (atm)  | T (K)     | Fit error |
|--------------------------------------------|------------|-------|---------|----------|-----------|-----------|
| $C_2H_3 + O_2 = CH_2CHOO^\bullet$          | $1.49E+15$ | -2.74 | -9657.  | 0.01 atm | 750–3000  | 15%       |
| $C_2H_3 + O_2 = CH_3CHOO^\bullet$          | $3.66E+10$ | -0.95 | -15810. | 1.0 atm  | 1000–3000 | 6%        |
| $C_2H_3 + O_2 = CH_2CHOO^\bullet$          | $1.73E+14$ | -1.65 | -12511. | 10.0 atm | 1000–3000 | 7%        |
| $C_2H_3 + O_2 = C^\bullet CHO + O$         | $9.63E+11$ | 0.07  | -413.   | 0.01 atm | 300–3000  | 13%       |
| $C_2H_3 + O_2 = C^\bullet CHO + O$         | $3.86E+12$ | -0.08 | 588.    | 1.0 atm  | 500–3000  | 15%       |
| $C_2H_3 + O_2 = C^\bullet CHO + O$         | $1.13E+11$ | 0.36  | 42.     | 10.0 atm | 750–3000  | 10%       |
| $C_2H_3 + O_2 = C_2H_2 + HO_2$ (concerted) | $8.20E+02$ | 2.42  | -3307.  | 0.01 atm | 500–3000  | 10%       |
| $C_2H_3 + O_2 = C_2H_2 + HO_2$ (concerted) | $2.23E+05$ | 1.76  | -1202.  | 1.0 atm  | 500–3000  | 15%       |
| $C_2H_3 + O_2 = C_2H_2 + HO_2$ (concerted) | $1.56E+03$ | 2.37  | -2451.  | 10.0 atm | 750–3000  | 9%        |
| $C_2H_3 + O_2 = CH_2O + HCO$ (4-mem. ring) | $1.36E+05$ | 1.53  | -294.   | 0.01 atm | 300–3000  | 9%        |
| $C_2H_3 + O_2 = CH_2O + HCO$ (4-mem. ring) | $5.77E+05$ | 1.37  | 549.    | 1.0 atm  | 500–3000  | 10%       |
| $C_2H_3 + O_2 = CH_2O + HCO$ (4-mem. ring) | $2.12E+10$ | 0.13  | 4669.   | 10.0 atm | 300–3000  | 14%       |
| $C_2H_3 + O_2 = C^\bullet CYCOO$           | $1.02E+11$ | -1.51 | -10122. | 0.01 atm | 750–3000  | 6%        |
| $C_2H_3 + O_2 = C^\bullet CYCOO$           | $3.95E+14$ | -1.93 | -8475.  | 1.0 atm  | 750–3000  | 8%        |
| $C_2H_3 + O_2 = C^\bullet CYCOO$           | $1.81E+20$ | -3.17 | -3243.  | 10.0 atm | 750–3000  | 12%       |
| $C_2H_3 + O_2 = C^\bullet HCHOOH$          | $6.43E+29$ | -7.20 | 324.    | 0.01 atm | 500–3000  | 4%        |
| $C_2H_3 + O_2 = C^\bullet HCHOOH$          | $5.86E+39$ | -9.33 | 9046.   | 1.0 atm  | 300–3000  | 10%       |
| $C_2H_3 + O_2 = C^\bullet HCHOOH$          | $2.41E+40$ | -9.17 | 12721.  | 10.0 atm | 500–3000  | 7%        |
| $C_2H_3 + O_2 = C_2H_2O + OH$              | $7.67E+16$ | -1.67 | 4379.   | 0.01 atm | 300–3000  | 13%       |
| $C_2H_3 + O_2 = C_2H_2O + OH$              | $6.47E+16$ | -1.64 | 4804.   | 1.0 atm  | 500–3000  | 12%       |
| $C_2H_3 + O_2 = C_2H_2O + OH$              | $5.52E+15$ | -1.32 | 4663.   | 10.0 atm | 750–3000  | 9%        |
| $C_2H_3 + O_2 = C_2H_2 + HO_2$ (2-step)    | $4.64E+07$ | 1.18  | 6158.   | 0.01 atm | 500–3000  | 9%        |
| $C_2H_3 + O_2 = C_2H_2 + HO_2$ (2-step)    | $1.40E+08$ | 1.05  | 6574.   | 1.0 atm  | 500–3000  | 10%       |
| $C_2H_3 + O_2 = C_2H_2 + HO_2$ (2-step)    | $4.16E+10$ | 0.37  | 8780.   | 10.0 atm | 500–3000  | 15%       |
| $C_2H_3 + O_2 = H_2C^\bullet OCHO$         | $4.07E-06$ | 2.71  | -22679. | 0.01 atm | 1000–3000 | 8%        |
| $C_2H_3 + O_2 = H_2C^\bullet OCHO$         | $1.23E-03$ | 2.58  | -22125. | 1.0 atm  | 1000–3000 | 8%        |
| $C_2H_3 + O_2 = H_2C^\bullet OCHO$         | $4.92E+00$ | 1.90  | -19038. | 10.0 atm | 1000–3000 | 10%       |
| $C_2H_3 + O_2 = CH_2O + HCO$ (3-mem. ring) | $3.54E+17$ | -1.73 | 798.    | 0.01 atm | 300–3000  | 13%       |
| $C_2H_3 + O_2 = CH_2O + HCO$ (3-mem. ring) | $4.19E+13$ | -0.63 | -1868.  | 1.0 atm  | 750–3000  | 8%        |

Table 3. (Continued).

| Reaction                                       | A        | n      | E      | P (atm)  | T (K)     | Fit error |
|------------------------------------------------|----------|--------|--------|----------|-----------|-----------|
| $C_2H_3 + O_2 = CH_2O + HCO$ (3-mem. ring)     | 2.69E+18 | -1.91  | 3259.  | 10.0 atm | 750–3000  | 13%       |
| $CH_3CHOO^\bullet = C_2H_3 + O_2$              | 3.81E+29 | -6.11  | 42300. | 0.01 atm | 750–3000  | 8%        |
| $CH_3CHOO^\bullet = C_2H_3 + O_2$              | 1.28E+33 | -6.52  | 43884. | 1.0 atm  | 750–3000  | 9%        |
| $CH_3CHOO^\bullet = C_2H_3 + O_2$              | 1.56E+38 | -7.60  | 48521. | 10.0 atm | 750–3000  | 13%       |
| $CH_3CHOO^\bullet = C^\bullet CHO + O$         | 5.98E+24 | -4.94  | 33513. | 0.01 atm | 750–3000  | 12%       |
| $CH_3CHOO^\bullet = C^\bullet CHO + O$         | 1.33E+24 | -4.20  | 31914. | 1.0 atm  | 1000–3000 | 6%        |
| $CH_3CHOO^\bullet = C^\bullet CHO + O$         | 2.52E+29 | -5.30  | 38429. | 10.0 atm | 1000–3000 | 7%        |
| $CH_3CHOO^\bullet = C_2H_2 + HO_2$ (concerted) | 2.15E+14 | -2.24  | 33011. | 0.01 atm | 750–3000  | 15%       |
| $CH_3CHOO^\bullet = C_2H_2 + HO_2$ (concerted) | 1.27E+10 | -0.56  | 27318. | 1.0 atm  | 1000–3000 | 6%        |
| $CH_3CHOO^\bullet = C_2H_2 + HO_2$ (concerted) | 4.03E+14 | -1.47  | 31756. | 10.0 atm | 1000–3000 | 8%        |
| $CH_3CHOO^\bullet = CH_2O + HCO$ (4-mem. ring) | 2.95E+15 | -2.84  | 36219. | 0.01 atm | 750–3000  | 13%       |
| $CH_3CHOO^\bullet = CH_2O + HCO$ (4-mem. ring) | 1.72E+19 | -3.31  | 38060. | 1.0 atm  | 750–3000  | 15%       |
| $CH_3CHOO^\bullet = CH_2O + HCO$ (4-mem. ring) | 5.85E+15 | -2.08  | 34673. | 10.0 atm | 1000–3000 | 6%        |
| $CH_3CHOO^\bullet = C.CYCOC$                   | 1.23E+41 | -10.10 | 29436. | 0.01 atm | 300–3000  | 6%        |
| $CH_3CHOO^\bullet = C.CYCOC$                   | 2.08E+42 | -9.68  | 32315. | 1.0 atm  | 300–3000  | 8%        |
| $CH_3CHOO^\bullet = C.CYCOC$                   | 1.04E+43 | -9.46  | 35368. | 10.0 atm | 500–3000  | 2%        |
| $CH_3CHOO^\bullet = C^\bullet HCHOH$           | 1.84E+29 | -7.43  | 33442. | 0.01 atm | 750–3000  | 5%        |
| $CH_3CHOO^\bullet = C^\bullet HCHOH$           | 5.37E+40 | -9.47  | 44551. | 1.0 atm  | 750–3000  | 9%        |
| $CH_3CHOO^\bullet = C^\bullet HCHOH$           | 6.95E+46 | -10.68 | 52358. | 10.0 atm | 500–3000  | 8%        |
| $CH_3CHOO^\bullet = C_2H_2O + OH$              | 4.14E+26 | -5.64  | 44561. | 0.01 atm | 750–3000  | 7%        |
| $CH_3CHOO^\bullet = C_2H_2O + OH$              | 4.09E+29 | -5.90  | 45664. | 1.0 atm  | 750–3000  | 8%        |
| $CH_3CHOO^\bullet = C_2H_2O + OH$              | 1.58E+34 | -6.85  | 50027. | 10.0 atm | 750–3000  | 12%       |
| $CH_3CHOO^\bullet = C_2H_2 + HO_2$ (2-step)    | 5.69E+13 | -1.91  | 45016. | 0.01 atm | 750–3000  | 10%       |
| $CH_3CHOO^\bullet = C_2H_2 + HO_2$ (2-step)    | 8.20E+15 | -1.95  | 45184. | 1.0 atm  | 750–3000  | 10%       |
| $CH_3CHOO^\bullet = C_2H_2 + HO_2$ (2-step)    | 1.24E+18 | -2.27  | 46446. | 10.0 atm | 750–3000  | 11%       |
| $CH_3CHOO^\bullet = H_2C^\bullet OCHO$         | 4.99E+14 | -4.07  | 11351. | 0.01 atm | 1000–3000 | 7%        |
| $CH_3CHOO^\bullet = H_2C^\bullet OCHO$         | 7.62E+24 | -5.66  | 22133. | 1.0 atm  | 1000–3000 | 9%        |
| $CH_3CHOO^\bullet = H_2C^\bullet OCHO$         | 1.30E+28 | -5.95  | 27598. | 10.0 atm | 1000–3000 | 9%        |
| $CH_3CHOO^\bullet = CH_2O + HCO$ (3-mem. ring) | 2.64E+36 | -8.28  | 33741. | 0.01 atm | 750–3000  | 11%       |
| $CH_3CHOO^\bullet = CH_2O + HCO$ (3-mem. ring) | 2.68E+45 | -10.09 | 44936. | 1.0 atm  | 500–3000  | 9%        |

|                                                                                   |        |        |          |           |     |
|-----------------------------------------------------------------------------------|--------|--------|----------|-----------|-----|
| $\text{CH}_3\text{CHOO}^\bullet = \text{CH}_3\text{O} + \text{HCO}$ (3-mem. ring) | -10.16 | 49738. | 10.0 atm | 500–3000  | 9%  |
| $\text{C.CYC} = \text{CH}_2\text{CHOO}^\bullet$                                   | -10.88 | 22258. | 0.01 atm | 300–3000  | 5%  |
| $\text{C.CYC} = \text{CH}_2\text{CHCOO}^\bullet$                                  | -10.18 | 24908. | 1.0 atm  | 300–3000  | 10% |
| $\text{C.CYC} = \text{CH}_2\text{CHOO}^\bullet$                                   | -10.01 | 28342. | 10.0 atm | 500–3000  | 2%  |
| $\text{C.CYC} = \text{C}_2\text{H}_3 + \text{O}_2$                                | -6.30  | 37204. | 0.01 atm | 500–3000  | 8%  |
| $\text{C.CYC} = \text{C}_2\text{H}_3 + \text{O}_2$                                | -7.30  | 40552. | 1.0 atm  | 500–3000  | 14% |
| $\text{C.CYC} = \text{C}_2\text{H}_3 + \text{O}_2$                                | -8.83  | 46505. | 10.0 atm | 300–3000  | 15% |
| $\text{C.CYC} = \text{C}^\bullet\text{CHO} + \text{O}$                            | -5.57  | 29849. | 0.01 atm | 500–3000  | 15% |
| $\text{C.CYC} = \text{C}^\bullet\text{CHO} + \text{O}$                            | -5.64  | 31774. | 1.0 atm  | 750–3000  | 9%  |
| $\text{C.CYC} = \text{C}^\bullet\text{CHO} + \text{O}$                            | -7.78  | 41405. | 10.0 atm | 500–3000  | 13% |
| $\text{C.CYC} = \text{C}_2\text{H}_2 + \text{HO}_2$ (concerted)                   | -1.52  | 25212. | 0.01 atm | 750–3000  | 7%  |
| $\text{C.CYC} = \text{C}_2\text{H}_2 + \text{HO}_2$ (concerted)                   | -2.16  | 27709. | 1.0 atm  | 750–3000  | 9%  |
| $\text{C.CYC} = \text{C}_2\text{H}_2 + \text{HO}_2$ (concerted)                   | -3.54  | 33783. | 10.0 atm | 750–3000  | 13% |
| $\text{C.CYC} = \text{CH}_3\text{O} + \text{HCO}$ (4-mem. ring)                   | -2.13  | 28449. | 0.01 atm | 750–3000  | 4%  |
| $\text{C.CYC} = \text{CH}_3\text{O} + \text{HCO}$ (4-mem. ring)                   | -2.61  | 30320. | 1.0 atm  | 750–3000  | 7%  |
| $\text{C.CYC} = \text{CH}_3\text{O} + \text{HCO}$ (4-mem. ring)                   | -3.91  | 35838. | 10.0 atm | 750–3000  | 10% |
| $\text{C.CYC} = \text{C}^\bullet\text{HCHO} + \text{H}$                           | -7.45  | 26834. | 0.01 atm | 750–3000  | 5%  |
| $\text{C.CYC} = \text{C}^\bullet\text{HCHO} + \text{H}$                           | -10.90 | 42071. | 1.0 atm  | 300–3000  | 11% |
| $\text{C.CYC} = \text{C}^\bullet\text{HCHO} + \text{H}$                           | -9.83  | 43923. | 10.0 atm | 300–3000  | 14% |
| $\text{C.CYC} = \text{C}_2\text{H}_2 + \text{OH}$                                 | -6.14  | 39571. | 0.01 atm | 500–3000  | 9%  |
| $\text{C.CYC} = \text{C}_2\text{H}_2 + \text{OH}$                                 | -7.03  | 42633. | 1.0 atm  | 500–3000  | 15% |
| $\text{C.CYC} = \text{C}_2\text{H}_2 + \text{OH}$                                 | -6.65  | 43230. | 10.0 atm | 750–3000  | 5%  |
| $\text{C.CYC} = \text{C}_2\text{H}_2 + \text{HO}_2$ (2-step)                      | -2.79  | 41555. | 0.01 atm | 500–3000  | 12% |
| $\text{C.CYC} = \text{C}_2\text{H}_2 + \text{HO}_2$ (2-step)                      | -2.98  | 42177. | 1.0 atm  | 500–3000  | 14% |
| $\text{C.CYC} = \text{C}_2\text{H}_2 + \text{HO}_2$ (2-step)                      | -1.98  | 39557. | 10.0 atm | 750–3000  | 4%  |
| $\text{C.CYC} = \text{H}_2\text{C}^\bullet\text{OCHO}$                            | -2.54  | 2508.  | 0.01 atm | 1000–3000 | 7%  |
| $\text{C.CYC} = \text{H}_2\text{C}^\bullet\text{OCHO}$                            | -4.03  | 12638. | 1.0 atm  | 1000–3000 | 9%  |
| $\text{C.CYC} = \text{H}_2\text{C}^\bullet\text{OCHO}$                            | -5.21  | 20841. | 10.0 atm | 750–3000  | 13% |
| $\text{C.CYC} = \text{CH}_3\text{O} + \text{HCO}$ (3-mem. ring)                   | -5.31  | 21231. | 0.01 atm | 750–3000  | 10% |
| $\text{C.CYC} = \text{CH}_3\text{O} + \text{HCO}$ (3-mem. ring)                   | -5.92  | 28670. | 1.0 atm  | 300–3000  | 12% |
| $\text{C.CYC} = \text{CH}_3\text{O} + \text{HCO}$ (3-mem. ring)                   | -5.25  | 30066. | 10.0 atm | 300–3000  | 7%  |
| $\text{C}^\bullet\text{HCHO} = \text{CH}_2\text{CHOO}^\bullet$                    | -8.52  | 8297.  | 0.01 atm | 750–3000  | 5%  |
| $\text{C}^\bullet\text{HCHO} = \text{CH}_2\text{CHOO}^\bullet$                    | -10.50 | 19237. | 1.0 atm  | 750–3000  | 8%  |

Table 3. (Continued).

| Reaction                                                                  | A        | n      | E      | P (atm)  | T (K)     | Fit error |
|---------------------------------------------------------------------------|----------|--------|--------|----------|-----------|-----------|
| $\text{C}^*\text{HCHOH} = \text{CH}_2\text{CHOO}^*$                       | 2.50E+48 | -11.77 | 27309. | 10.0 atm | 500-3000  | 8%        |
| $\text{C}^*\text{HCHOH} = \text{C}_2\text{H}_3 + \text{O}_2$              | 2.91E+42 | -10.72 | 24682. | 0.01 atm | 300-3000  | 9%        |
| $\text{C}^*\text{HCHOH} = \text{C}_2\text{H}_3 + \text{O}_2$              | 5.11E+48 | -11.82 | 30813. | 1.0 atm  | 300-3000  | 8%        |
| $\text{C}^*\text{HCHOH} = \text{C}_2\text{H}_3 + \text{O}_2$              | 8.60E+51 | -12.38 | 36188. | 10.0 atm | 500-3000  | 3%        |
| $\text{C}^*\text{HCHOH} = \text{C}^*\text{CHO} + \text{O}$                | 4.45E+39 | -9.85  | 21618. | 0.01 atm | 300-3000  | 12%       |
| $\text{C}^*\text{HCHOH} = \text{C}^*\text{CHO} + \text{O}$                | 1.32E+47 | -11.28 | 30159. | 1.0 atm  | 500-3000  | 4%        |
| $\text{C}^*\text{HCHOH} = \text{C}^*\text{CHO} + \text{O}$                | 8.40E+46 | -10.93 | 33587. | 10.0 atm | 500-3000  | 4%        |
| $\text{C}^*\text{HCHOH} = \text{C}_2\text{H}_2 + \text{HO}_2$ (concerted) | 9.37E+29 | -7.54  | 18526. | 0.01 atm | 500-3000  | 5%        |
| $\text{C}^*\text{HCHOH} = \text{C}_2\text{H}_2 + \text{HO}_2$ (concerted) | 4.50E+39 | -9.58  | 27452. | 1.0 atm  | 300-3000  | 10%       |
| $\text{C}^*\text{HCHOH} = \text{C}_2\text{H}_2 + \text{HO}_2$ (concerted) | 3.28E+40 | -9.50  | 31426. | 10.0 atm | 500-3000  | 5%        |
| $\text{C}^*\text{HCHOH} = \text{CH}_2\text{O} + \text{HCO}$ (4-mem. ring) | 5.72E+29 | -7.80  | 20119. | 0.01 atm | 500-3000  | 5%        |
| $\text{C}^*\text{HCHOH} = \text{CH}_2\text{O} + \text{HCO}$ (4-mem. ring) | 1.56E+39 | -9.79  | 28569. | 1.0 atm  | 300-3000  | 8%        |
| $\text{C}^*\text{HCHOH} = \text{CH}_2\text{O} + \text{HCO}$ (4-mem. ring) | 2.79E+40 | -9.80  | 32692. | 10.0 atm | 500-3000  | 4%        |
| $\text{C}^*\text{HCHOH} = \text{C.CYCOO}$                                 | 2.04E+33 | -9.19  | 13145. | 0.01 atm | 500-3000  | 15%       |
| $\text{C}^*\text{HCHOH} = \text{C.CYCOO}$                                 | 1.70E+44 | -11.07 | 23647. | 1.0 atm  | 300-3000  | 11%       |
| $\text{C}^*\text{HCHOH} = \text{C.CYCOO}$                                 | 2.11E+42 | -10.03 | 25643. | 10.0 atm | 500-3000  | 13%       |
| $\text{C}^*\text{HCHOH} = \text{C}_2\text{H}_2\text{O} + \text{OH}$       | 1.87E+27 | -5.75  | 21337. | 0.01 atm | 500-3000  | 5%        |
| $\text{C}^*\text{HCHOH} = \text{C}_2\text{H}_2\text{O} + \text{OH}$       | 4.21E+30 | -6.12  | 22612. | 1.0 atm  | 500-3000  | 6%        |
| $\text{C}^*\text{HCHOH} = \text{C}_2\text{H}_2\text{O} + \text{OH}$       | 2.81E+36 | -7.43  | 26906. | 10.0 atm | 300-3000  | 15%       |
| $\text{C}^*\text{HCHOH} = \text{C}_2\text{H}_2 + \text{HO}_2$ (2-step)    | 2.69E+17 | -2.76  | 23877. | 0.01 atm | 500-3000  | 5%        |
| $\text{C}^*\text{HCHOH} = \text{C}_2\text{H}_2 + \text{HO}_2$ (2-step)    | 3.36E+19 | -2.79  | 23963. | 1.0 atm  | 500-3000  | 5%        |
| $\text{C}^*\text{HCHOH} = \text{C}_2\text{H}_2 + \text{HO}_2$ (2-step)    | 1.64E+21 | -2.97  | 24591. | 10.0 atm | 500-3000  | 6%        |
| $\text{C}^*\text{HCHOH} = \text{H}_2\text{C}^*\text{OCHO}$                | 7.38E+26 | -8.00  | 7584.  | 0.01 atm | 750-3000  | 7%        |
| $\text{C}^*\text{HCHOH} = \text{H}_2\text{C}^*\text{OCHO}$                | 4.28E+38 | -10.08 | 18343. | 1.0 atm  | 750-3000  | 13%       |
| $\text{C}^*\text{HCHOH} = \text{H}_2\text{C}^*\text{OCHO}$                | 2.87E+44 | -11.14 | 26138. | 10.0 atm | 750-3000  | 14%       |
| $\text{C}^*\text{HCHOH} = \text{CH}_2\text{O} + \text{HCO}$ (3-mem. ring) | 6.33E+41 | -10.41 | 21936. | 0.01 atm | 300-3000  | 10%       |
| $\text{C}^*\text{HCHOH} = \text{CH}_2\text{O} + \text{HCO}$ (3-mem. ring) | 3.84E+49 | -11.93 | 31257. | 1.0 atm  | 500-3000  | 3%        |
| $\text{C}^*\text{HCHOH} = \text{CH}_2\text{O} + \text{HCO}$ (3-mem. ring) | 3.38E+52 | -12.47 | 37523. | 10.0 atm | 500-3000  | 2%        |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{CH}_2\text{CHOO}^*$              | 5.04E-03 | -0.54  | 4892.  | 0.01 atm | 1000-3000 | 7%        |

|                                                                                    |          |       |         |          |           |     |
|------------------------------------------------------------------------------------|----------|-------|---------|----------|-----------|-----|
| $\text{H}_2\text{C}^*\text{OCHO} = \text{CH}_2\text{CHOO}^*$                       | 5.30E+01 | -0.55 | 4909.   | 1.0 atm  | 1000-3000 | 7%  |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{CH}_2\text{CHOO}^*$                       | 5.59E+03 | -0.56 | 4927.   | 10.0 atm | 1000-3000 | 7%  |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}_3\text{H}_3 + \text{O}_2$              | 1.55E+17 | -3.36 | 104272. | 0.01 atm | 750-3000  | 13% |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}_3\text{H}_3 + \text{O}_2$              | 4.01E+20 | -3.74 | 105712. | 1.0 atm  | 750-3000  | 15% |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}_3\text{H}_3 + \text{O}_2$              | 9.26E+16 | -2.47 | 101908. | 10.0 atm | 1000-3000 | 6%  |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}^*\text{CHO} + \text{O}$                | 3.66E-01 | 1.25  | 85536.  | 0.01 atm | 1000-3000 | 8%  |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}^*\text{CHO} + \text{O}$                | 8.41E+03 | 0.63  | 88302.  | 1.0 atm  | 1000-3000 | 9%  |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}^*\text{CHO} + \text{O}$                | 1.42E+10 | -0.74 | 94972.  | 10.0 atm | 1000-3000 | 11% |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}_3\text{H}_2 + \text{HO}_2$ (concerted) | 4.21E-09 | 3.29  | 85744.  | 0.01 atm | 1000-3000 | 8%  |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}_3\text{H}_2 + \text{HO}_2$ (concerted) | 2.36E-06 | 3.09  | 86609.  | 1.0 atm  | 1000-3000 | 9%  |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}_3\text{H}_2 + \text{HO}_2$ (concerted) | 5.21E-02 | 2.21  | 90619.  | 10.0 atm | 1000-3000 | 11% |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{CH}_2\text{O} + \text{HCO}$ (4-mem. ring) | 1.94E-06 | 2.29  | 90018.  | 0.01 atm | 1000-3000 | 7%  |
| $\text{H}_2\text{O}^*\text{OCHO} = \text{CH}_2\text{O} + \text{HCO}$ (4-mem. ring) | 7.05E-04 | 2.14  | 90666.  | 1.0 atm  | 1000-3000 | 7%  |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{CH}_2\text{O} + \text{HCO}$ (4-mem. ring) | 4.86E+00 | 1.39  | 94044.  | 10.0 atm | 1000-3000 | 9%  |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}_3\text{CYCOO}$                         | 1.58E+06 | -3.02 | 16540.  | 0.01 atm | 750-3000  | 14% |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}_3\text{CYCOO}$                         | 1.93E+10 | -3.04 | 16596.  | 1.0 atm  | 750-3000  | 13% |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}_3\text{CYCOO}$                         | 2.46E+12 | -3.07 | 16663.  | 10.0 atm | 750-3000  | 13% |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}^*\text{HCHOOH}$                        | 1.42E+11 | -3.36 | 18291.  | 0.01 atm | 750-3000  | 10% |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}^*\text{HCHOOH}$                        | 1.42E+15 | -3.36 | 18291.  | 1.0 atm  | 750-3000  | 10% |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}^*\text{HCHOOH}$                        | 1.42E+17 | -3.36 | 18291.  | 10.0 atm | 750-3000  | 10% |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}_3\text{H}_2\text{O} + \text{OH}$       | 1.89E+16 | -3.54 | 106375. | 0.01 atm | 750-3000  | 15% |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}_3\text{H}_2\text{O} + \text{OH}$       | 3.70E+11 | -1.73 | 100148. | 1.0 atm  | 1000-3000 | 6%  |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}_3\text{H}_2\text{O} + \text{OH}$       | 7.58E+14 | -2.33 | 103261. | 10.0 atm | 1000-3000 | 7%  |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}_3\text{H}_2 + \text{HO}_2$ (2-step)    | 1.48E-03 | 1.92  | 100385. | 0.01 atm | 1000-3000 | 6%  |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}_3\text{H}_2 + \text{HO}_2$ (2-step)    | 1.79E-01 | 1.90  | 100485. | 1.0 atm  | 1000-3000 | 6%  |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{C}_3\text{H}_2 + \text{HO}_2$ (2-step)    | 9.18E+00 | 1.71  | 101352. | 10.0 atm | 1000-3000 | 7%  |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{CH}_2\text{O} + \text{HCO}$ (3-mem. ring) | 4.06E+21 | -4.16 | 22003.  | 0.01 atm | 1000-3000 | 9%  |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{CH}_2\text{O} + \text{HCO}$ (3-mem. ring) | 3.97E+39 | -8.36 | 42128.  | 1.0 atm  | 500-3000  | 15% |
| $\text{H}_2\text{C}^*\text{OCHO} = \text{CH}_2\text{O} + \text{HCO}$ (3-mem. ring) | 9.11E+33 | -6.46 | 41058.  | 10.0 atm | 500-3000  | 12% |

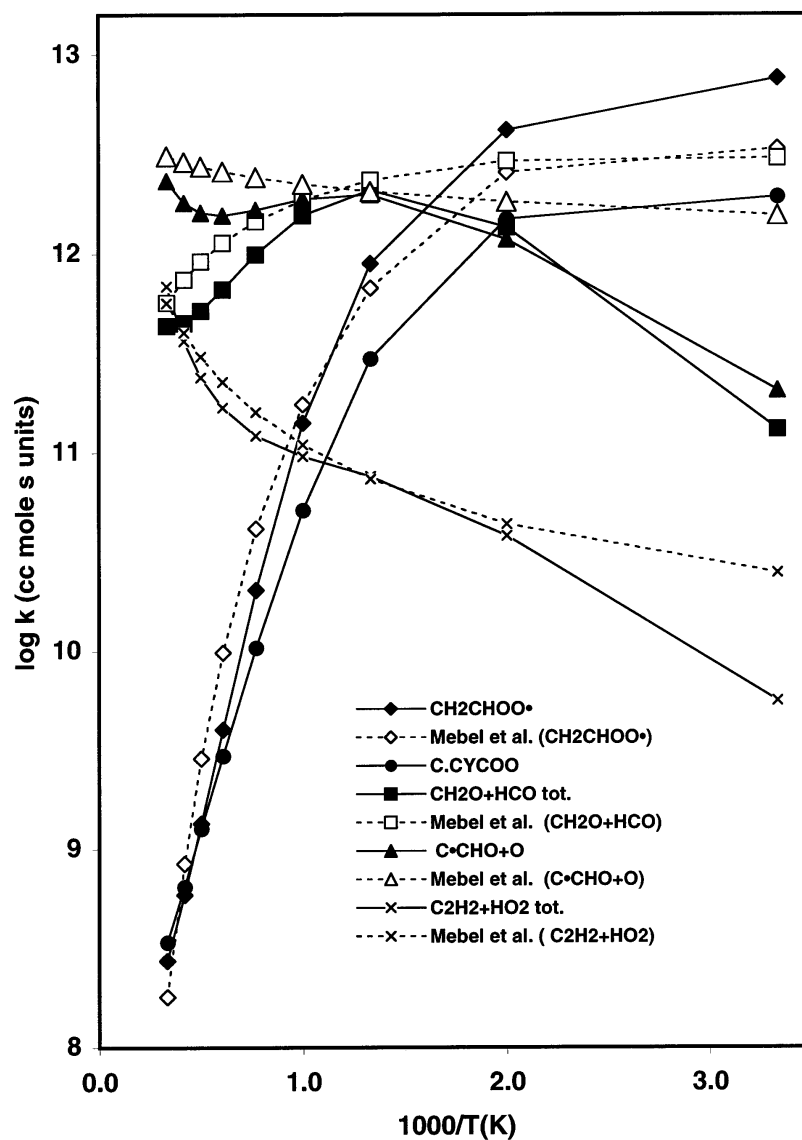
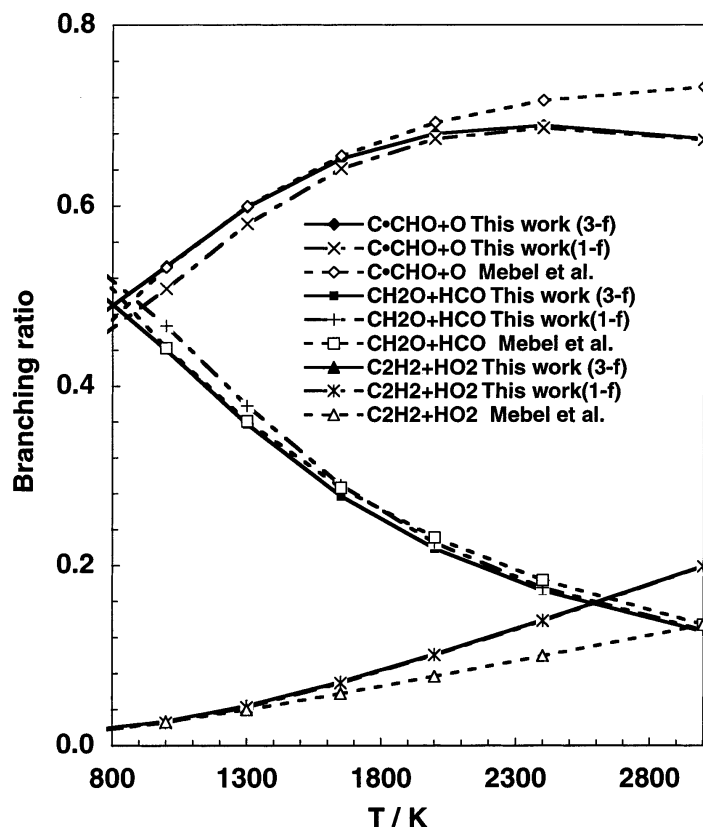


Fig. 7. Comparison of the CHEMACT3F results for the rate coefficients for the various channels for the reaction of vinyl radical with molecular oxygen to those predicted using a multichannel RRKM analysis by Mebel *et al.* [20]. Both calculations are for 1 atm  $\text{N}_2$ .



**Fig. 8.** Comparison of both the CHEMACT3F and single geometric mean frequency results for the high temperature branching ratios for the reaction of vinyl radical with molecular oxygen to those predicted using a multichannel RRKM analysis by Mebel *et al.* [20].

## Conclusions

CHEMACT3F allows predictions of the temperature and pressure dependence of both chemically-activated reactions and the subsequent reactions of stabilized adducts. The input parameters required for the calculations are generally available from macroscopic parameters and does not require information about detailed transition state properties. This code incorporates a variety of improvements over that described earlier [5]. A three-frequency model has been implemented; the code has been generalized to allow multiple wells, each of which can be interconnected in many ways, and the collisional stabilization model has also been improved. The code was used to predict the temperature- and pressure-dependent rate coefficients for the

addition of vinyl radicals to molecular oxygen. These calculations confirm earlier results that the vinoxy + O channel is dominant under combustion conditions. The results are consistent with RRKM results using the same input conditions. This approach provides a means to obtain reasonably accurate predictions of the rate coefficients that are required in many detailed mechanisms. The major advantage is the ability to provide reliable rate coefficients for many complex systems where detailed information about the transition states is not available. It is also shown that a simpler 1-frequency model appears adequate for high temperature conditions.

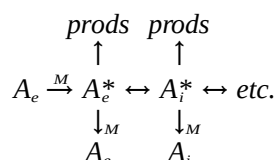
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## Appendix A: A compatible dissociation model

In a typical chemical activation reaction, pressure-dependent unimolecular dissociation rates are required to account for further reactions of stabilized adducts. For completeness, an analysis compatible with the chemical activation system presented above is summarized here. The dissociation problem begins by assuming stabilized adducts which are activated via collisions to an energy where they can undergo dissociation to products;



This clearly is a similar problem to that of chemical activation. However for each system, multiple calculations must be performed, each allowing a different isomer to be the starting point.

The pressure-dependent dissociation rate constants are defined through,

$$\frac{d[A_i]}{dt} = k^{stab}[A_e], \quad (\text{A1})$$

$$\frac{d[prod]}{dt} = k^{prod}[A_e]. \quad (\text{A2})$$

As before, we write an expression giving the rate of change of the population of each isomer  $i$ , with energy  $E$ ,

$$\begin{aligned}
 \frac{d[AC_i(E)]}{dt} = & k_i^s(T)[M]f'(E,T)\delta_{i,e} + \sum_j (IS^T)_{ij} AC_j(E) \\
 & - AC_i(E) \{ \sum_j IS_{ij} + \sum_m PD_{im} + k_i^s(T)[M] \},
 \end{aligned} \quad (\text{A3})$$

where  $f'(E,T)$  denotes the equilibrium  $E$  distribution of the entrance isomer,

$$f'(E,T) = \varrho_i(E, \Delta E; T) \exp(-E/kT). \quad (\text{A4})$$

This is identical with Eq. (3), except we have replaced the source term  $k^{inp}[R][R']f(E,T)\delta_{i,e}$  with the new source term  $k_i^s(T)[M]f'(E,T)\delta_{i,e}$ , which expresses the rate into level  $E$  of isomer  $A_e$  due to collisional activation. This new source term can easily be derived through a detailed balance analysis of the rates of collisional activation and stabilization. By applying the stea-

dy-state assumption to Eq. (A3) the following is obtained,

$$M_{i,j} \left\{ \frac{AC_j^{ss}(E)}{[M]} \right\} = -\delta_{i,e} k_i^s(T) [M] f'(E, T), \quad (A5)$$

where the matrix  $M_{i,j}$  is identical to that of Eq. (8). Eq. (A4) is solved at each  $E$  for the vector  $AC_j^{ss}(E)/[M]$ , and the integrated rate constants are then found via,

$$k_i^{stab} = \sum_E k_i^s(T) [M] \left\{ \frac{AC_i^{ss}(E)}{[M]} \right\}, \quad (A6)$$

$$k_m^{prod} = \sum_E k_{im}^s(E, \Delta E) \left\{ \frac{AC_i^{ss}(E)}{[M]} \right\}. \quad (A7)$$

The low pressure limiting cases can be analogously derived,

$$\frac{d[A_i]}{dt} = k_{lp,0}^{stab} [A_e] [M] + k_{lp,1}^{stab} [A_e] [M]^2, \quad (A8)$$

$$\frac{d[prod]}{dt} = k_{lp}^{prod} [A_e] [M], \quad (A9)$$

as can the high pressure limiting cases,

$$\frac{d[A_i]}{dt} = \frac{k_{hp}^{stab} [A_e]}{[M]^{(d-2)}}, \quad (A10)$$

$$\frac{d[prod]}{dt} = \frac{k_{hp}^{prod} [A_e]}{[M]^{(d-1)}}. \quad (A11)$$

Here the limiting rate constants are independent of pressure, and  $d$  gives the depth of the isomer immediately preceding the product channel relative to the dissociating isomer  $A_e$  (which is assigned a value  $d = 1$ ). The limiting cases are obtained using the identical high and low pressure forms of  $M_{i,j}$  as in the chemical activation case, and with the appropriate factorization of  $[M]$  from Eqs. (A5) and (A6).

The peculiar form for the low pressure limit in the stabilization case arises when a barrier to isomerization is less than any exit channel to products. In this case the matrix  $M_{i,j}$  has a different solution in the low pressure limit. This can be seen by taking the system of equations in Eq. (A5) and replacing the equation corresponding to the input well ( $i = e$ ) by the sum of all the equations. The right hand side is unchanged, but on the left hand side, the coefficients are replaced with the sum of all exit and stabilization channels for each well at that  $E$ . If an exit channel exists at this  $E$ , this equation will be dominated by the exit channels, and when  $[M]$  is taken to zero, the solution of Eq. (A5) will be proportional to the  $[M]$  of the source term. However, if all exit channels are zero, the factor of  $[M]$  will occur in

each term of the ( $i = e$ ) equation, and these cancel with the  $[M]$  of the source. For isomerization channels with such a low barrier, the sum over  $E$  will be composed of terms where there are no exit channels and, at higher  $E$ , of terms where there are such channels. The low pressure limit must thus be expressed as the sum of expressions of two different powers of  $P$ . For all normal product channel, terms only contribute for  $E$  above an exit barrier, and thus this situation never arises (likewise, in the chemical activation calculation, all  $E$ 's are taken above the input channel which also serves as an exit channel).

Finally the generalized Lindemann  $F$  factors are obtained through,

$$Pr^{stab} = \frac{k_{lp,0}^{stab}[M]^{(d-1)} + k_{lp,1}^{stab}[M]^d}{k_{hp}^{stab}}, \quad (A12)$$

$$Pr^{prod} = \frac{k_{lp}^{prod}[M]^d}{k_{hp}^{prod}}, \quad (A13)$$

$$k^{stab} = \frac{k_{hp}^{stab}}{[M]^{(d-2)}} \left( \frac{Pr^{stab}}{1 + Pr^{stab}} \right) F, \quad (A14)$$

$$k^{prod} = \frac{k_{hp}^{prod}}{[M]^{(d-1)}} \left( \frac{Pr^{prod}}{1 + Pr^{prod}} \right) F. \quad (A15)$$

## Appendix B: n-Frequency model

We wish to specify the distribution function  $N(k)$ , which corresponds to the total number of states of available energy  $E = kh\nu$  where  $\nu$  corresponds to the frequency we wish to use to index our system. In the harmonic oscillator model, this function  $N(k)$  is used in both the numerator and denominator of the differential rate functions. First, we define the distribution function,

$$N'_{if}(E, i) = \frac{I(k + s_i)}{I(k + 1)I(s_i)}, \quad k = \text{int}[E/(h\nu_i)], \quad (B1)$$

which is the number of ways to distribute energy  $E$  in a system of  $s_i$  degenerate oscillators with frequency  $\nu_i$ . In the case that the well is parameterized by a single frequency, we simply let  $N(k) = N'_{if}(E = k_1 h \nu_1, 1)$ . In the 2-frequency case, we must distribute energy  $E$  among two distinct sets of frequencies,  $\nu_1$  and  $\nu_2$ , (with respective degeneracies  $s_1$  and  $s_2$ ) and we must account for all combinations of  $k_1$  and  $k_2$  that give total energy  $E$ . Here, we have

$$N(k) = N'_{2f}[E = (k_1 + 1/2) h \nu_1] \quad (B2)$$

where,

$$N'_{2f}(E) = \sum_{k_2=0, E/(h\nu_2)} N'_{1f}(E_2, 2) N'_{1f}(E_{\text{residual}} = E - E_2, 1) . \quad (\text{B3})$$

Now we sum over possibilities of  $k_2$  from 0 to a maximum of  $\text{int}[E/(h\nu_2)]$ , while  $k_1$  is simultaneously adjusted so that the total energy remains constant. Again, the frequency  $\nu_1$  corresponds to that used to index the well in the dissociation or chemical activation calculation. The factor of 1/2 shifts the value of energy passed so that it corresponds not to level  $k$  but to the band defined by  $k$ .

This concept can be trivially extended. For the 3-frequency case, we let

$$N(k) = N'_{3f}[E = (k_1 + 1/2) h\nu_1] \quad (\text{B4})$$

where,

$$N'_{3f}(E) = \sum_{k_3=0, E/(h\nu_3)} N'_{1f}(E_3, 3) N'_{2f}(E_{\text{residual}} = E - E_3) , \quad (\text{B5})$$

and use Eq. (B3) to obtain  $N'_{2f}(E)$ . Generalizing for  $n$  frequencies, we obtain a set of recursive relations,

$$N'_{nf}(E) = \sum_{k_n=0, E/(h\nu_n)} N'_{1f}(E_n, n) N'_{(n-1)f}(E_{\text{residual}} = E - E_n) . \quad (\text{B6})$$

For a particular model employing  $n$  distinct frequencies, Eq. (B6) would be invoked through the initial substitution,

$$N(k) = N'_{nf}[E = (k_1 + 1/2) h\nu_1] . \quad (\text{B7})$$

The ordering of the recursive components is immaterial, so long as the innermost element corresponds to  $\nu_1$ .

## Incorporation of 1-D rotors

The inclusion of  $n$  1-D internal rotors in the density of states is straightforward (this is normally not needed in the RRK analysis since the  $n$ -frequency model fit to the specific heat captures the hindered internal rotors adequately – however it is useful for modeling the active rotor of a transition state, should the need arise). In this case, the same prescription is followed, with Eq. (B1) used for all  $N$  but the innermost  $N$  (that corresponding to  $\nu_1$ ). This last function is replaced by the convolution of Eq. (B1) with the function  $[\pi^{n/2}/\Gamma(n/2)k^{n/2-1}]$  on interval  $k$ . For  $n = 1$ , the factor  $k^{-1/2}$  is replaced by unity at  $k = 0$ . The factor involving the moments of inertia in the rotational density of states can be neglected because it is similarly neglected in the rotational partition function [i.e.  $Q_{\text{rot}} = (\pi RT)n/2$ ]. The coarse graining of the convolution used here (based on  $\nu_1$ ) results in noticeable error for  $n > 1$ , but seems acceptable for the typical case of  $n = 1$ .